UNIVERSITY OF GEORGIA

Infectious Disease Molecular Epidemiology

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- Professor, Department of Infectious Diseases, Department of Epidemiology and Biostatistics, Institute of Bioinformatics.
- Lead the Applied Molecular Epidemiology Research Group
- Focus on RNA viruses (Influenza, Coronaviruses, Dengue, HIV, etc)
  - Endemic and emerging pathogens
- Use pathogen genomic data to learn about evolutionary processes that are hard to observe

Our Challenge:

How do we better use genomic data to inform public health

- Better integration of genomic technologies in public health agencies
  - Laboratory technicians
  - Bioinformatic computational biology support
  - Computer infrastructure
  - Integrated data management
  - Method development and validation

How prepared are we to use pathogen genomics in public health?

Audience survey + U.S. training landscape

Who is in the room?

Choose the role that best describes your current work.

- **A** Public health laboratory / microbiology
- **B** Epidemiology / surveillance / outbreak response
- **C** Bioinformatics / data science
- **D** Leadership, program management, or other
- **E** My role crosses two or more of these areas

How often do you use genomic evidence?

In your current public health role…

- **A** Never or almost never
- **B** I receive genomic results, but rarely interpret them
- **C** I interpret results with support from others
- **D** I routinely integrate genomic and epidemiologic data
- **E** I generate or analyze genomic data

Where are you least confident?

Rate your confidence: 1 = low | 5 = high

Interpret a phylogenetic tree	1 2 3 4 5
Distinguish genetic similarity from transmission	1 2 3 4 5
Recognize sampling bias and missingness	1 2 3 4 5
Combine genomic, case, and exposure data	1 2 3 4 5
Translate evidence into a public health recommendation	1 2 3 4 5

How were you trained for this work?

Select every route that applies.

- **A** Formal university course, concentration, or certificate
- **B** Short course, workshop, or professional training
- **C** On-the-job training or mentoring
- **D** Self-directed learning
- **E** No structured training yet

“Molecular epidemiology” usually means something else

Most formal U.S. programs grew from three established traditions.

BIOMARKERS

Exposure, susceptibility, and molecular mechanisms

Environmental health and cancer epidemiology

HOST GENETICS

Human genetic variation and disease risk

Genetic epidemiology and precision health

LAB EPIDEMIOLOGY

Microbiology linked with epidemiologic methods

Hospital and infectious disease epidemiology

Pathogen evolution + genomic surveillance + outbreak decision-making are rarely combined into a formal credential.

Formal credentials exist—but few center pathogen genomics

Institution	Formal offering	Primary orientation	Pathogen genomics
Michigan SPH	MPH concentration	Hospital + laboratory epi	Relevant, not organizing focus
Emory Rollins	Graduate certificate	Host genetics + biomarkers	Limited
Columbia Mailman	MPH certificate	Exposure + disease mechanisms	Limited
University of Georgia CPH	PhD research track	Molecular + genetic epi	Possible; mentor dependent
UAB SPH	PhD concentration area	Individualized molecular epi	Variable
Yale SPH	MPH in microbial diseases	Infectious disease epi	Strong component; not a named credential

Landscape scan of named U.S. school-of-public-health offerings; course availability and curricula change over time.

The formal training gap is pathogen genomics

DEGREE ENVIRONMENT **Yale EMD MPH**	Genomic epidemiology is available within a broader microbial-disease degree.
TARGETED RESEARCH TRAINING **Pittsburgh SAPPHGenE**	Genuine public health genomic epidemiology training, but restricted to a funded partnership.
WORKFORCE EDUCATION **Harvard / CDC PGCoE**	Free, focused courses for public health practice—but not an academic credential.
LOCAL COURSES + MENTORSHIP **Multiple universities**	Expertise exists, but learners assemble their own pathway.

Expertise is present. A coherent, practice-facing training pathway is not.

The missing space is integration for action

GENOMIC EVIDENCE

- Pathogen biology + evolution
- Sequencing + bioinformatics literacy
- **Phylogenetic inference**
- Sampling bias + uncertainty

PUBLIC HEALTH DECISIONS

- Outbreak linkage assessment
- Surveillance design
- Integration with case + exposure data
- **Public health recommendations**

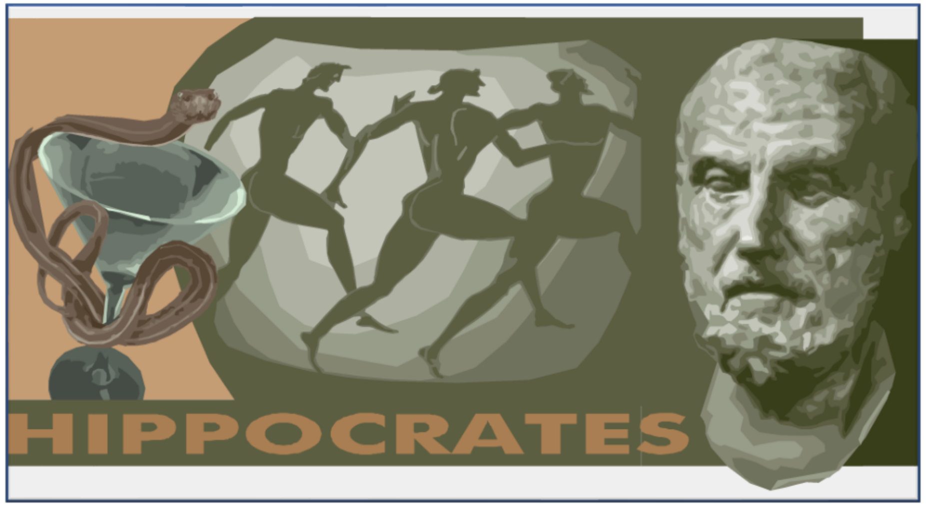
This program trains the interface:

How to judge genomic evidence—and use it responsibly in surveillance and outbreak response.

What is Epidemiology?

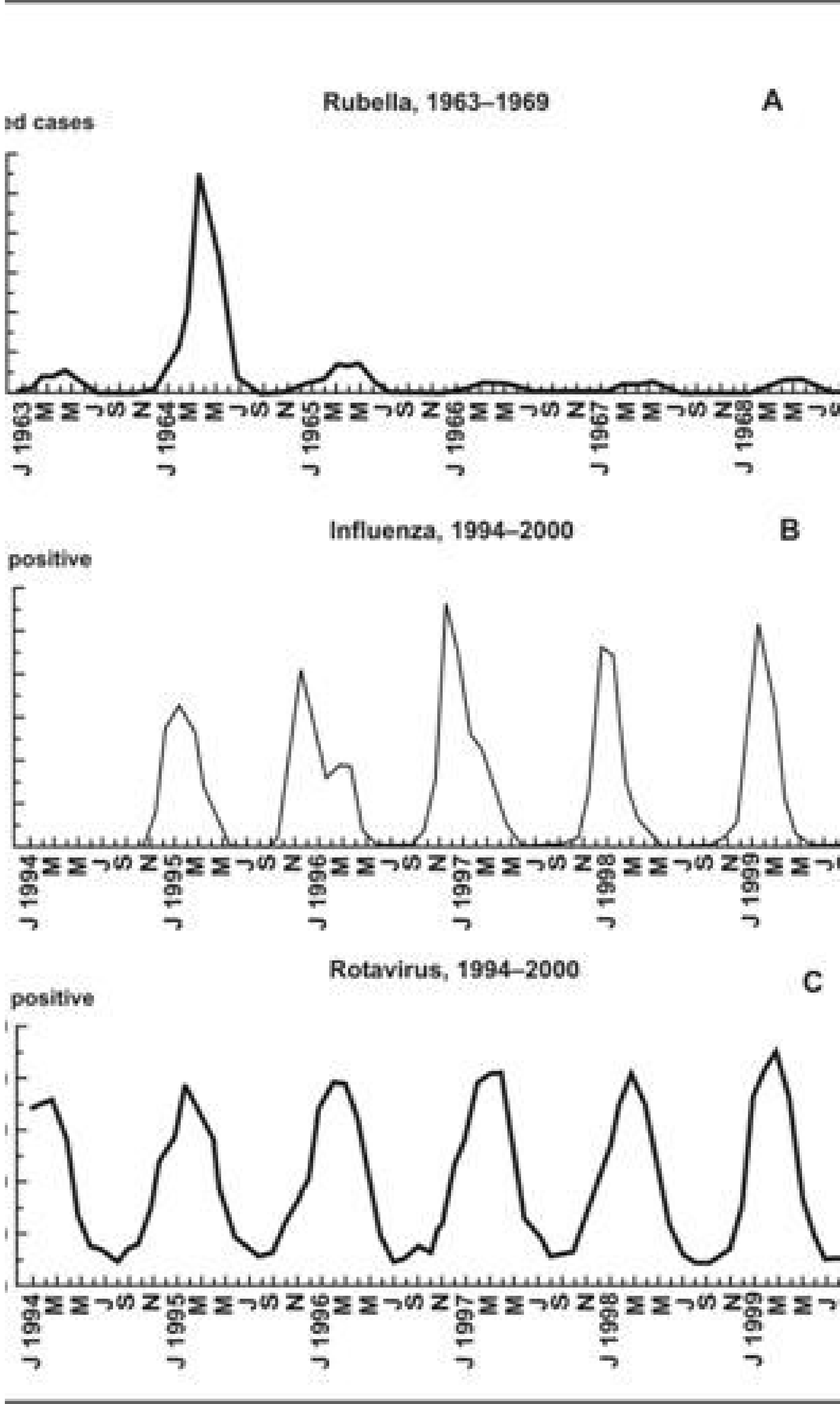
Etymology

- epi (upon) + demos (people) + logos (study)
- Epidemiology
- “the study of what is upon the people”



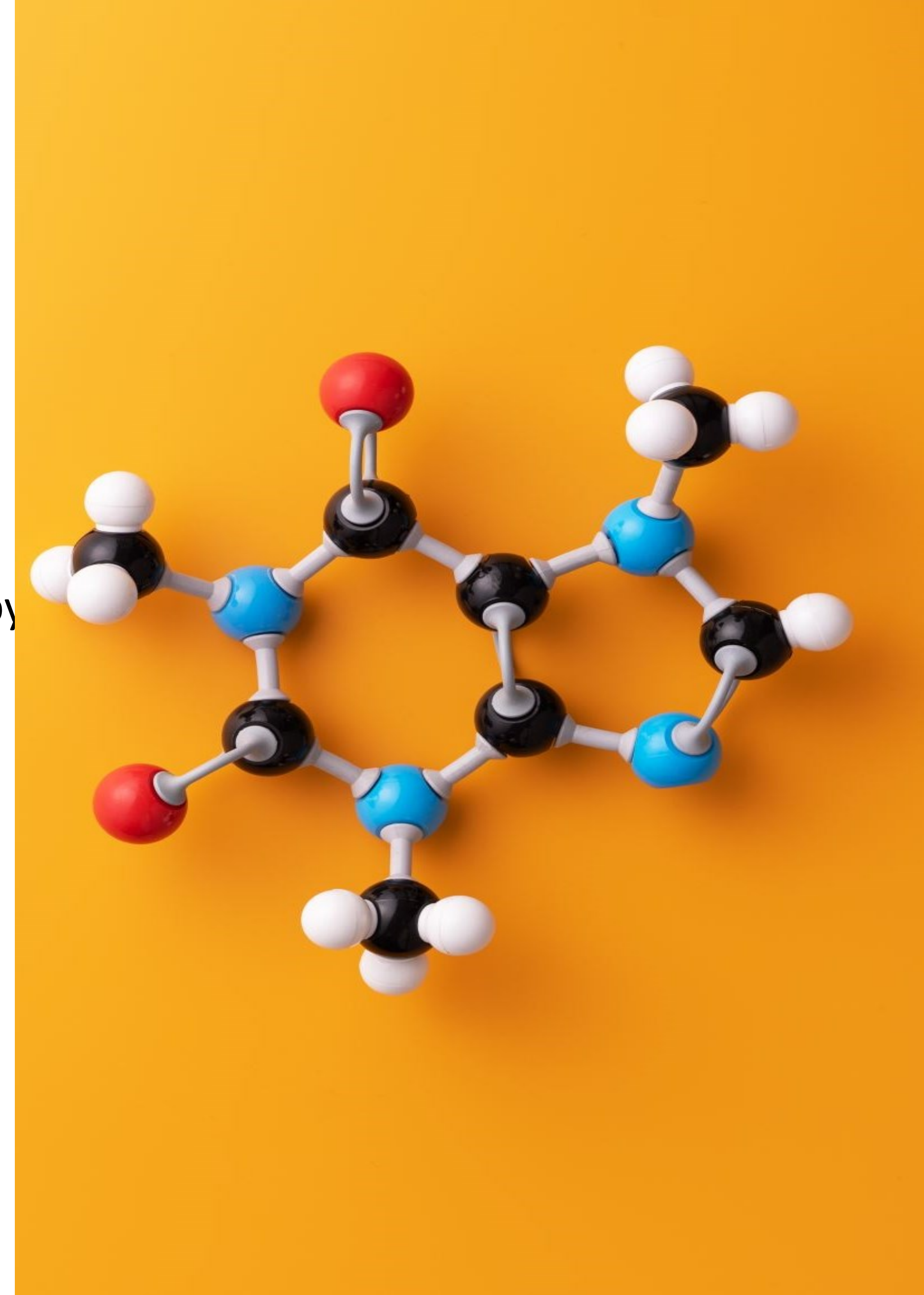
The Five W's of Epidemiology

- What = health issue of concern
  Who = person
  Where = place
  When = time
  Why/how = causes, risk factors, modes of transmission



What is molecular epidemiology?

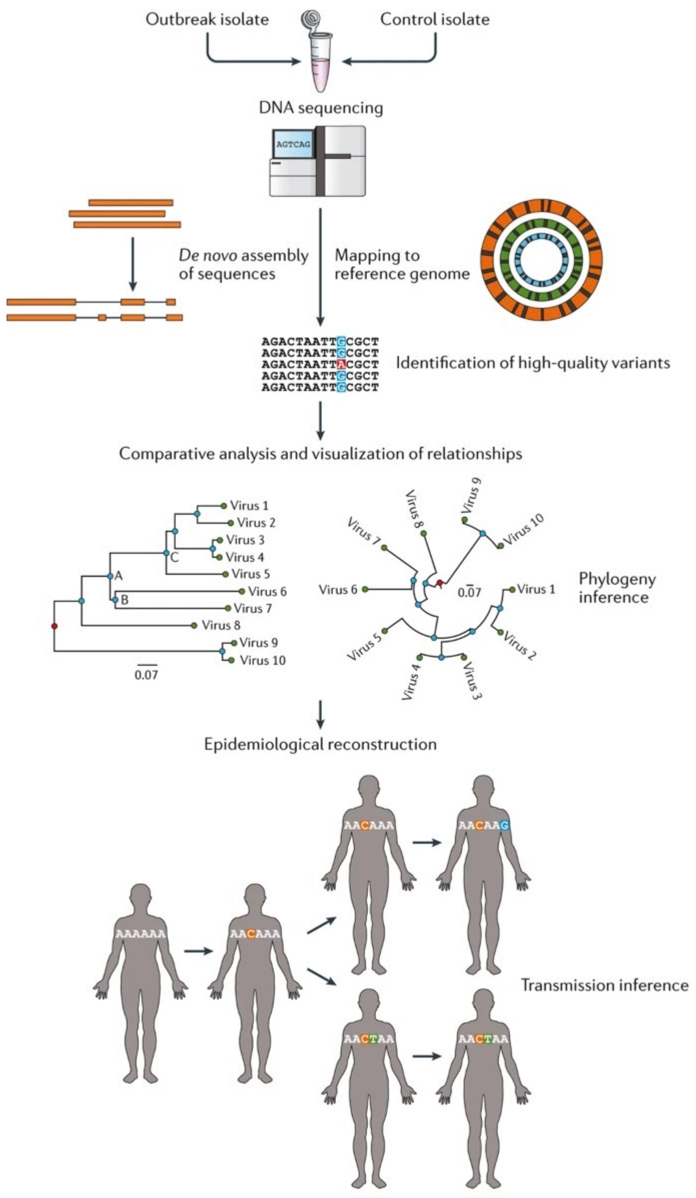
- Molecular epidemiology is a branch of epidemiology and medical science developed by merging molecular biology into epidemiological studies.



Infectious Disease Molecular Epidemiology

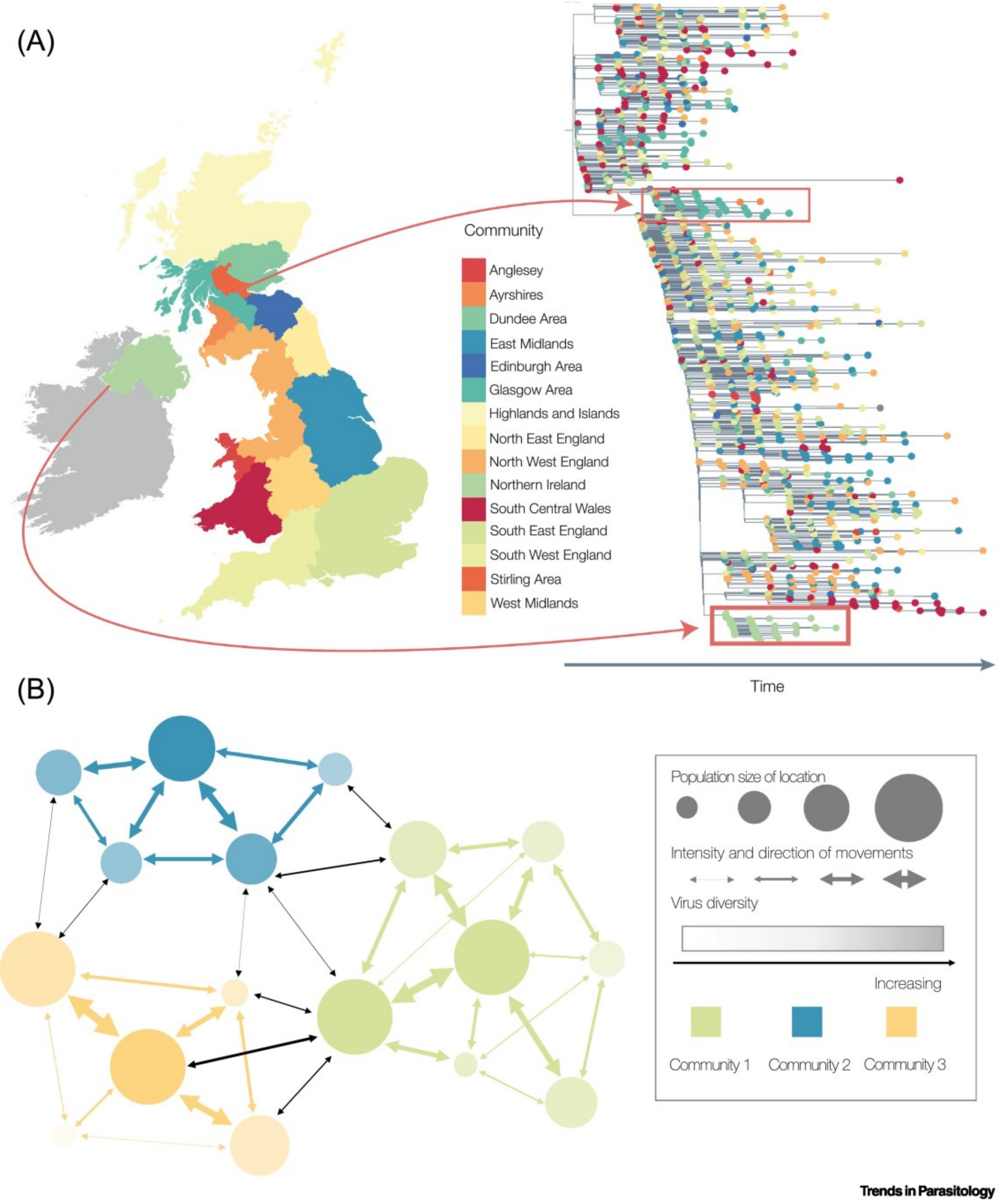
- aims to identify infectious agents, unravel the source, identify their reservoirs and trace circulation pattern and transmission pattern using (genomic sequence) data

Real-time genomic epidemiology of infectious disease outbreaks

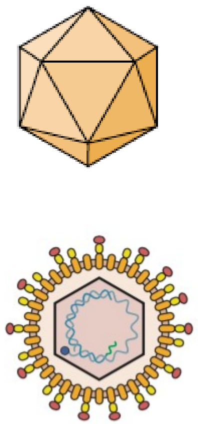
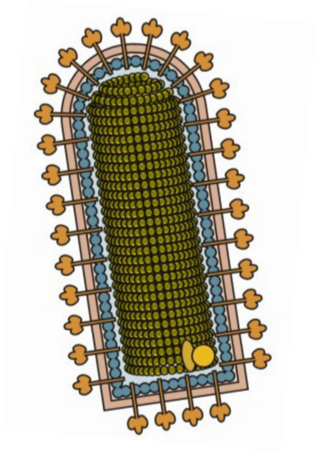


Nature Reviews | Genetics

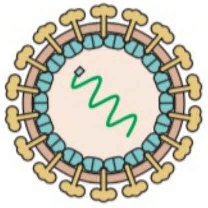
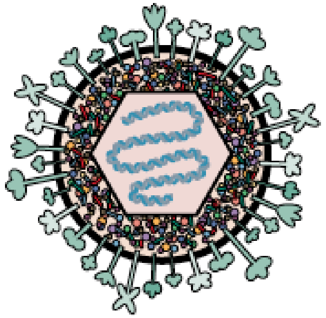
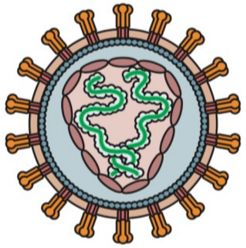
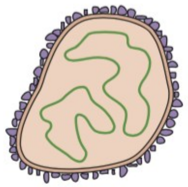
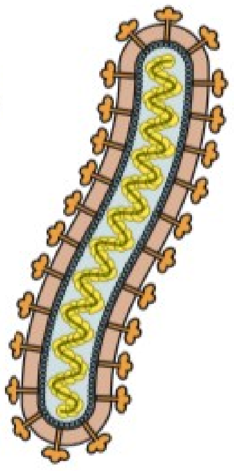
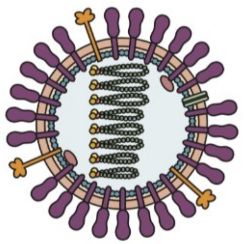
Spatial Epidemiology across scales



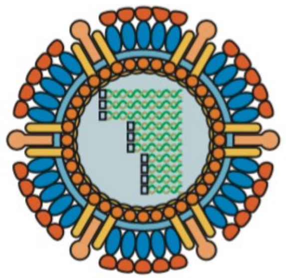
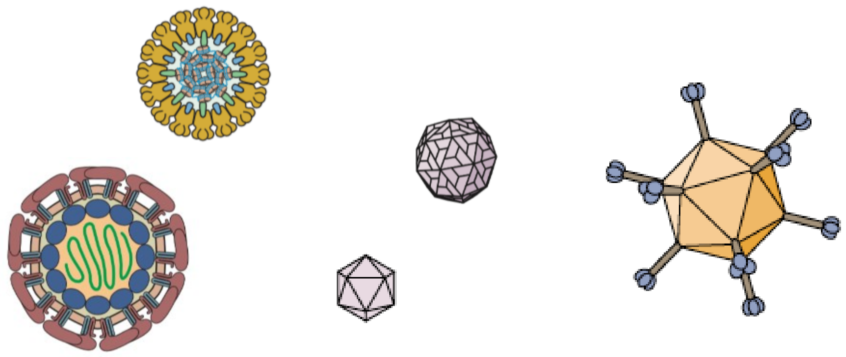
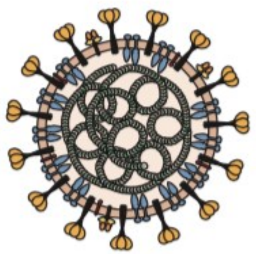
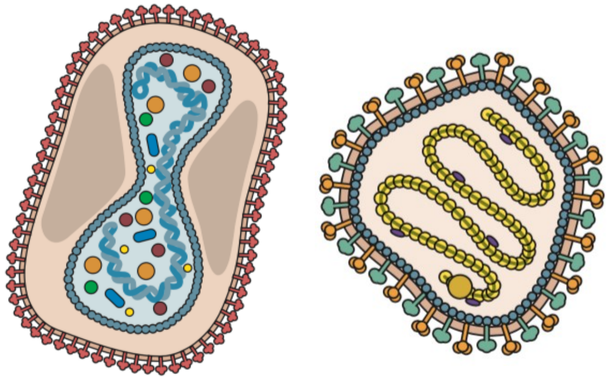
Trends in Parasitology 2021 371038-1049DOI: (10.1016/j.pt.2021.08.007)

What is a virus?

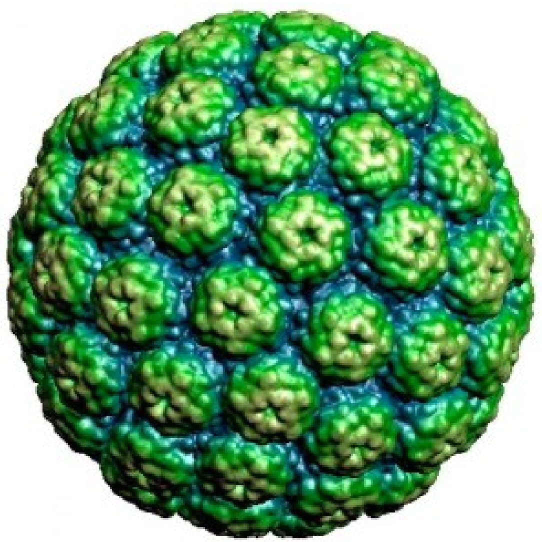







An *infectious, obligate intracellular parasite comprising genetic material* (DNA or RNA) surrounded by a protein coat and/or a membrane

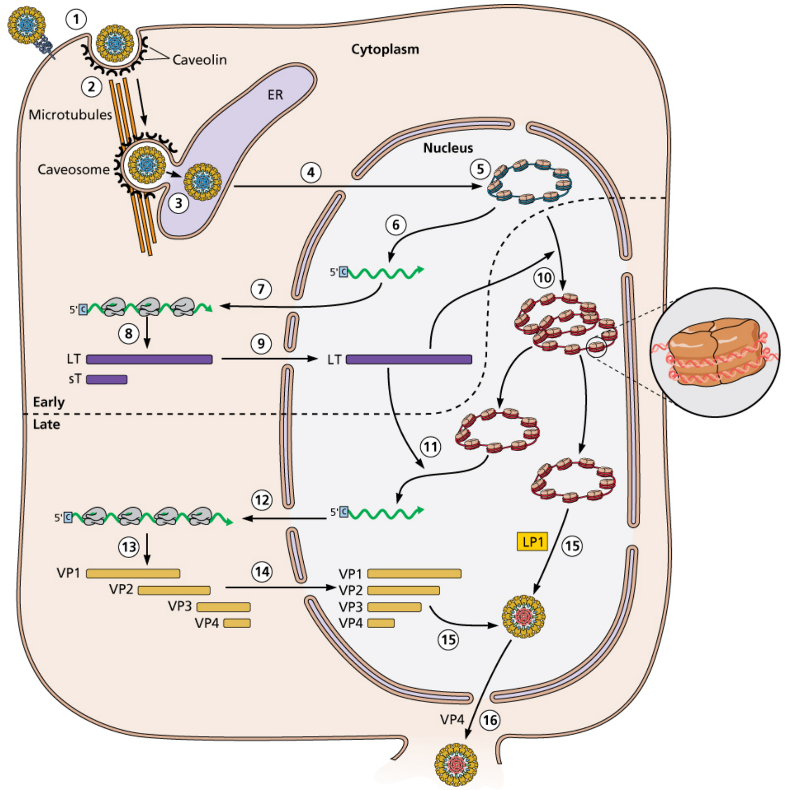





The virus and the virion

A virus is an organism with two phases



virion



infected cell

The before story – viral disease

- Infectious disease has played a major role in human evolution and societal development.
- Microbiology, clinical ID and epidemiology have common origins in medicine
- The apparent separation of infectious disease epidemiology from medicine is recent and superficial

Epidemics through history

430
B.C.

Smallpox is caused by the variola virus, which spreads through skin-to-skin contact or contact with bodily fluids. It can also be spread through the air.

In 430 B.C., smallpox killed more than 30,000 people in Athens, Greece, reducing the city's population by at least 20%.

Epidemics through history

1519

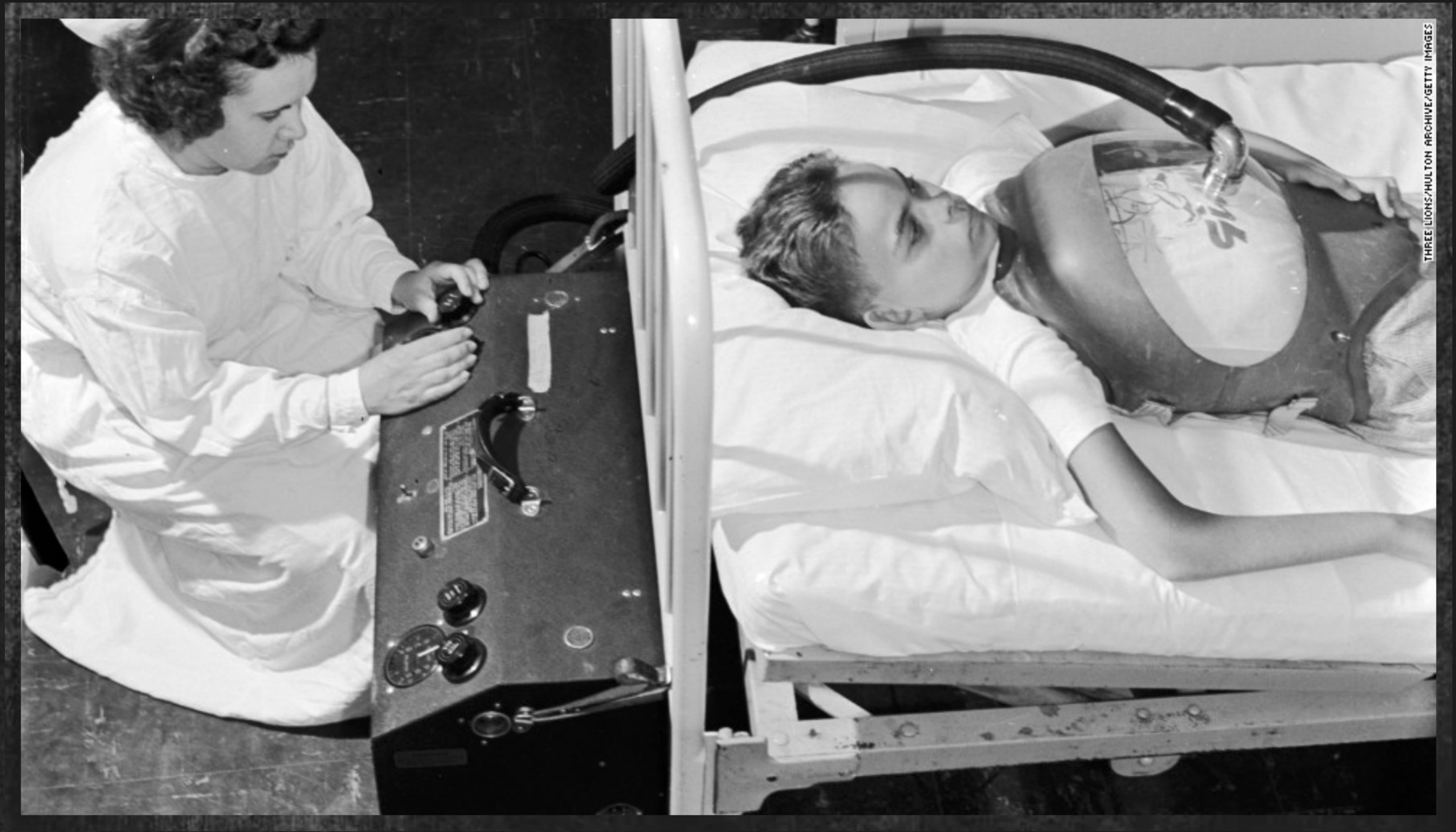
There were approximately 25 million people living in what is now called Mexico when Hernando Cortes arrived in 1519. A smallpox epidemic killed between 5 and 8 million of the native population in the following two years. Over the next century, less than 2 million would survive this and other communicable diseases brought by European explorers.

Epidemics through history

1918

The great flu pandemic of 1918 and 1919 is estimated to have killed between 30 million and 50 million people worldwide. Among them were 675,000 Americans.

Epidemics through history



1952

Polio peaked in the US. Nearly 60,000 children were infected and more than 3,000 died. Three years later vaccination began to prevent the communicable disease.

Epidemics through history

1984

In 1984, scientists identified the human immunodeficiency virus, or HIV, as the cause of AIDS. That same year the deadly disease killed more than 5,500 people in the United States. Today more than 35 million people around the world are living with an HIV infection. More than 25 million people have died of AIDS since the first cases were reported.

Epidemics through history

2003

Severe Acute Respiratory Syndrome, better known as SARS, was first identified in 2003 in China, though the first case is believed to have occurred in November 2002. By July more than 8,000 cases and 774 deaths had been reported.

The Modern Era of Molecular Epidemiology begins!....

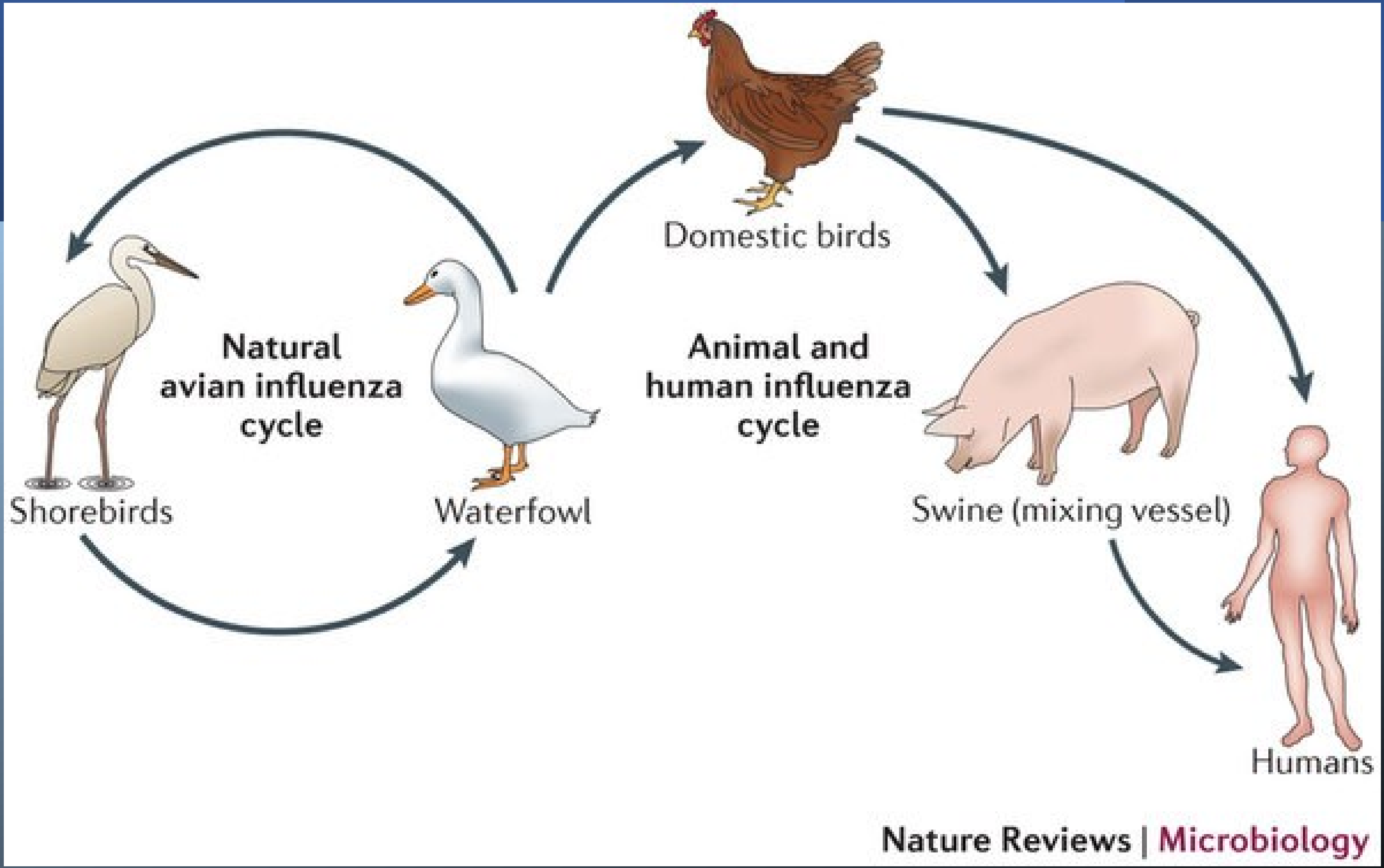
2009

The global H1N1 flu pandemic may have killed as many as 575,000 people, though only 18,500 deaths were confirmed. The H1N1 virus is a type of swine flu, which is a respiratory disease of pigs caused by the type A influenza virus.

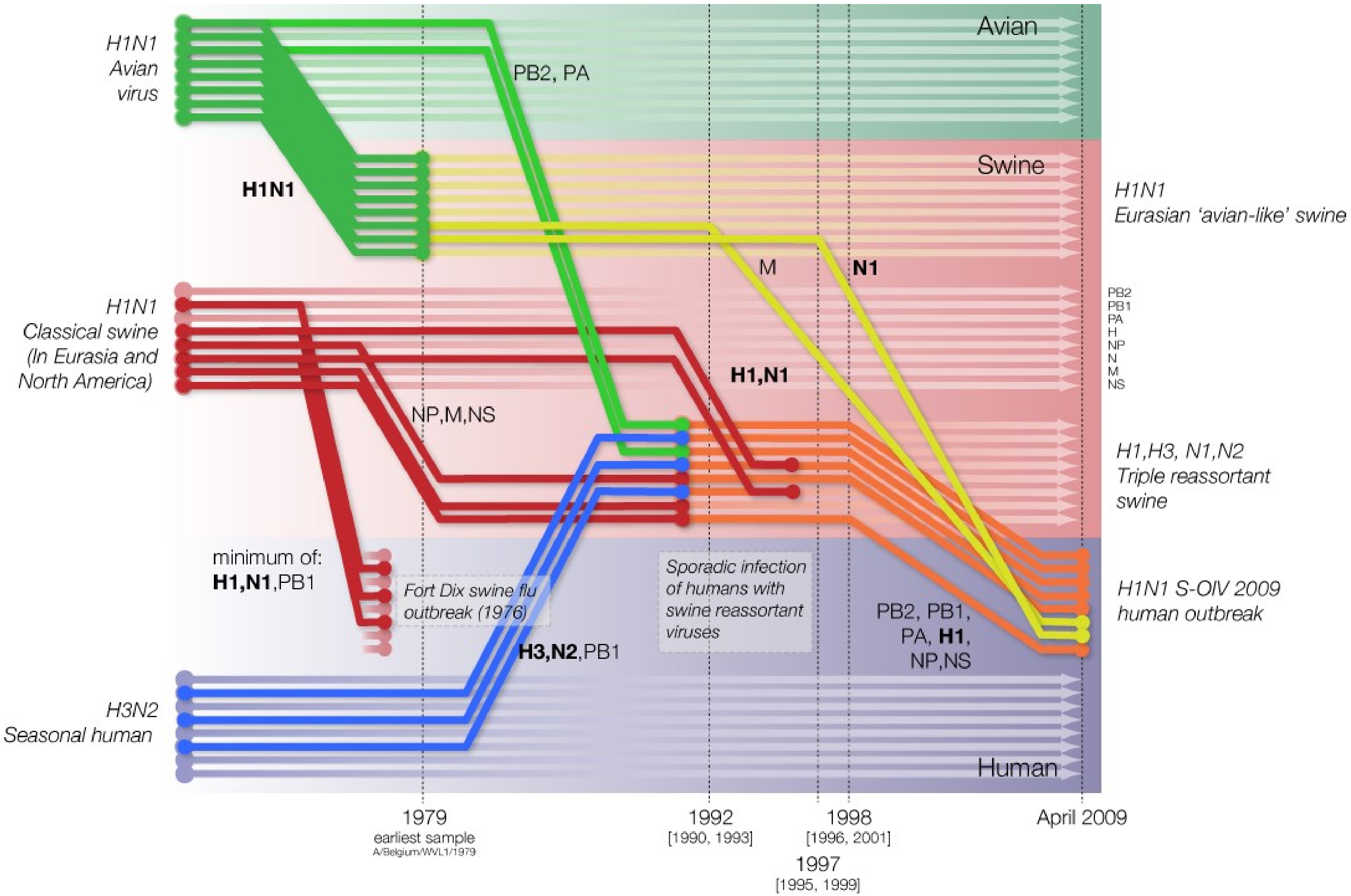
Swine Origin Influenza Virus (H1N1pdm)

Timeline

- April 15 and April 17, 2009, novel swine-origin influenza A (H1N1) virus (S-OIV) was identified in two epidemiologically unlinked patients in the United States
- The same strain of the virus was identified in Mexico, Canada
- Spread across the globe in weeks
- Global network of scientists was made aware of the new threat April 23/24th
- First sequences of the novel influenza virus was available April 25th


Domestic birds
Natural
avian influenza
cycle
Animal and
human influenza
cycle
Shorebirds
Waterfowl
Swine (mixing vessel)
Humans
Nature Reviews | Microbiology

H1N1pdm


Avian
Swine
Human
H1N1
Avian
virus
H1N1
PB2, PA
H1N1
Eurasian 'avian-like' swine
M
N1
H1N1
Classical swine
(In Eurasia and
North America)
PB2
PB1
PA
H
NP
N
M
NS
H1,N1
NP,M,NS
H1,H3, N1,N2
Triple reassortant
swine
minimum of:
H1,N1,PB1
Fort Dix swine flu
outbreak (1976)
Sporadic infection
of humans with
swine reassortant
viruses
PB2, PB1,
PA, H1,
NP,NS
H1N1 S-OIV 2009
human outbreak
H3,N2,PB1
H3N2
Seasonal human
1979
earliest sample
A/Belgium/WVL1/1979
1992
[1990, 1993]
1998
[1996, 2001]
1997
[1995, 1999]
April 2009

History of Infectious Disease

Pre-Germ Theory Era

- Era of plagues
  - Early epidemiology
- Discovery of micro-organisms

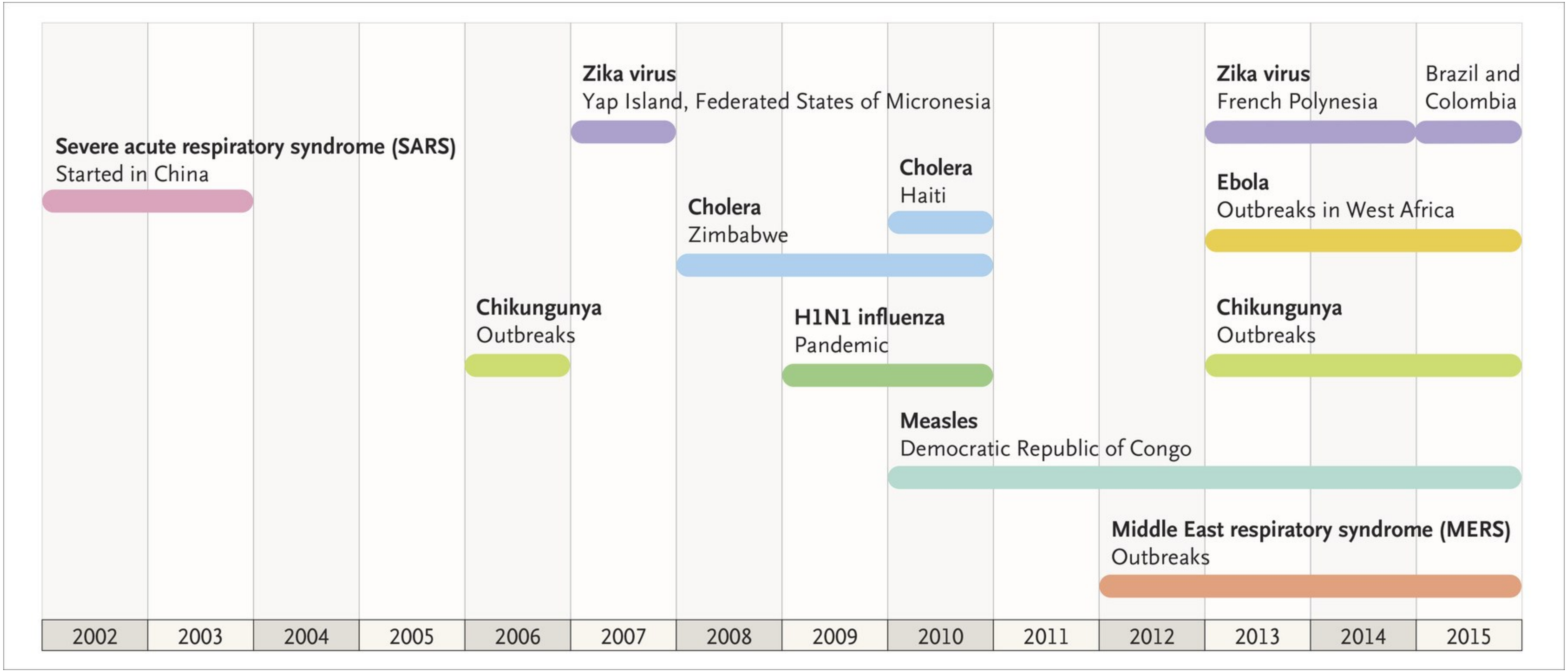
19^{th} Century – Discovery and Development

Post Germ Theory Era

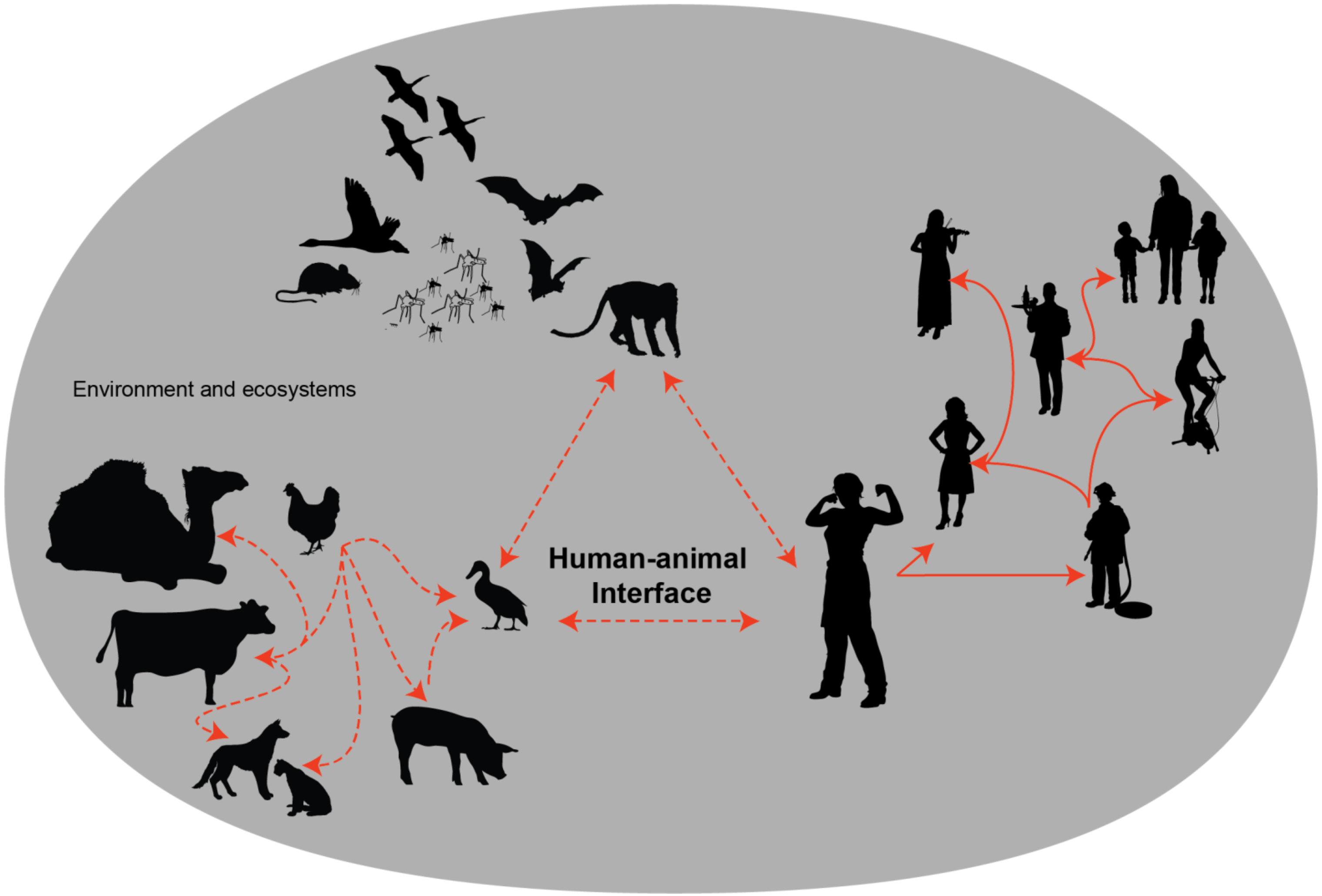
Modern Era (Needs a better name)

- Open Source
- Molecular Approaches
- Fast Paced

Epidemics in the genomic age



Comparative genomics of pathogens and populations across biological scales



Evolutionary processes give rise to diversity

- There are four basic mechanisms by which biological evolution takes place. These include mutation, migration, genetic drift, and natural selection.

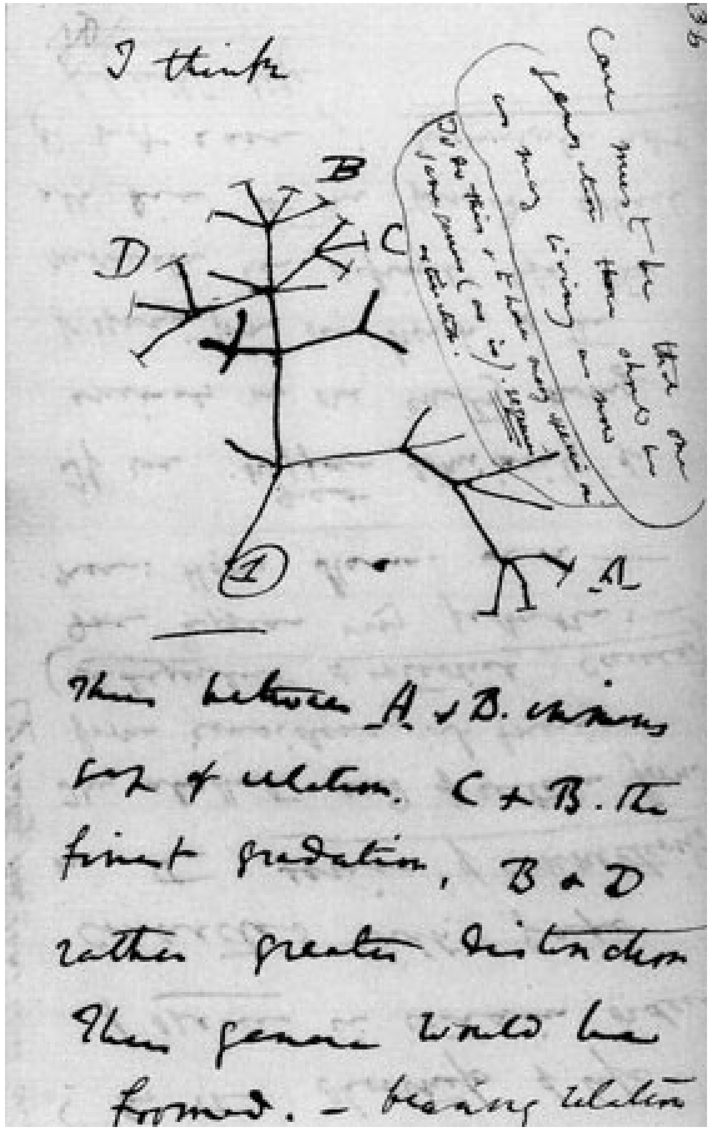
- When we sequence a population of viruses we are observing genetic diversity resulting from those processes.

A phylogenetic tree

is an evolutionary hypothesis showing the history and relationships (common descent) of the observed (genetic) diversity

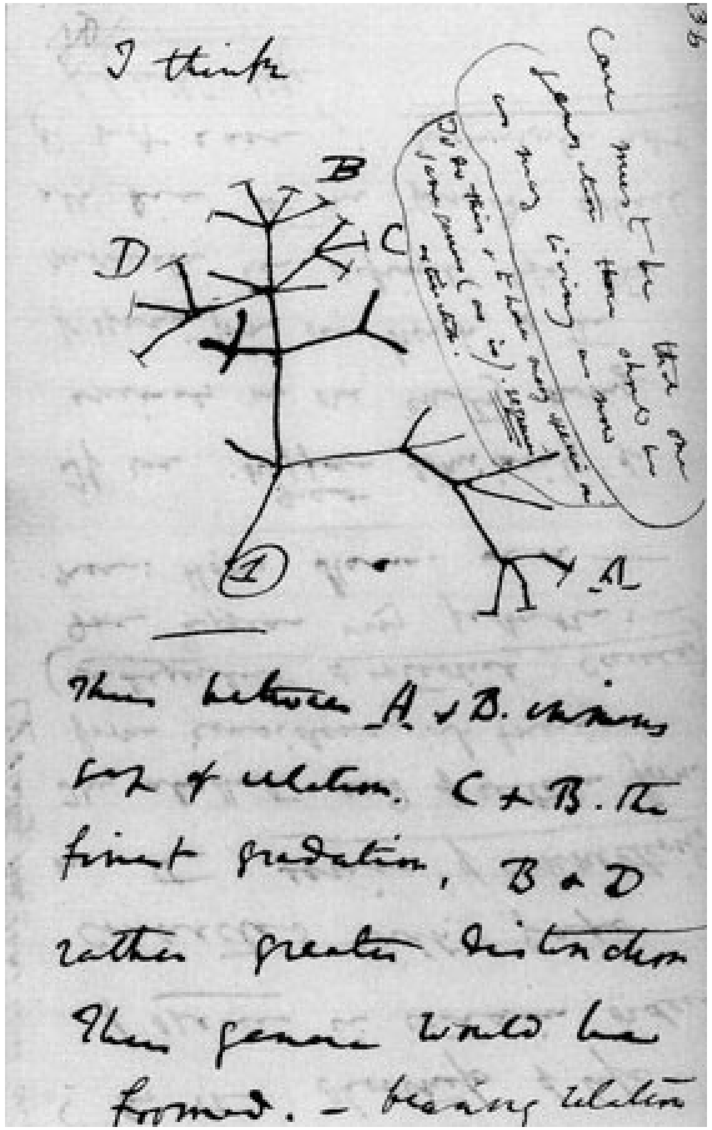
Ecology, Evolution and Epidemiology of viruses

- Darwin and Wallace formulate a theory of evolution by common descent
- 1859 Darwin publishes *The Origin of Species*
- Modern Synthesis (1920-40)

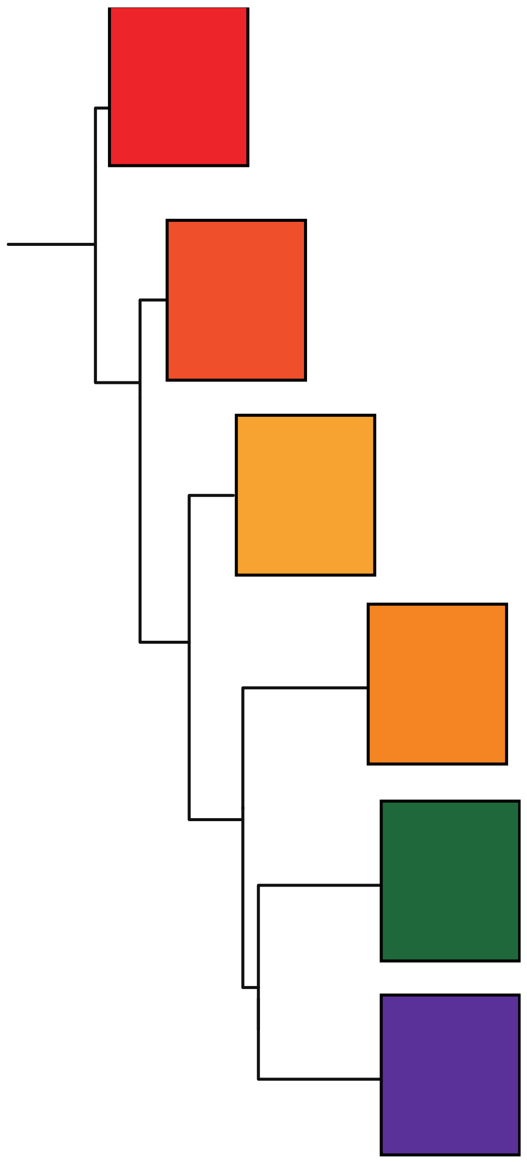


Ecology, Evolution and Epidemiology of viruses

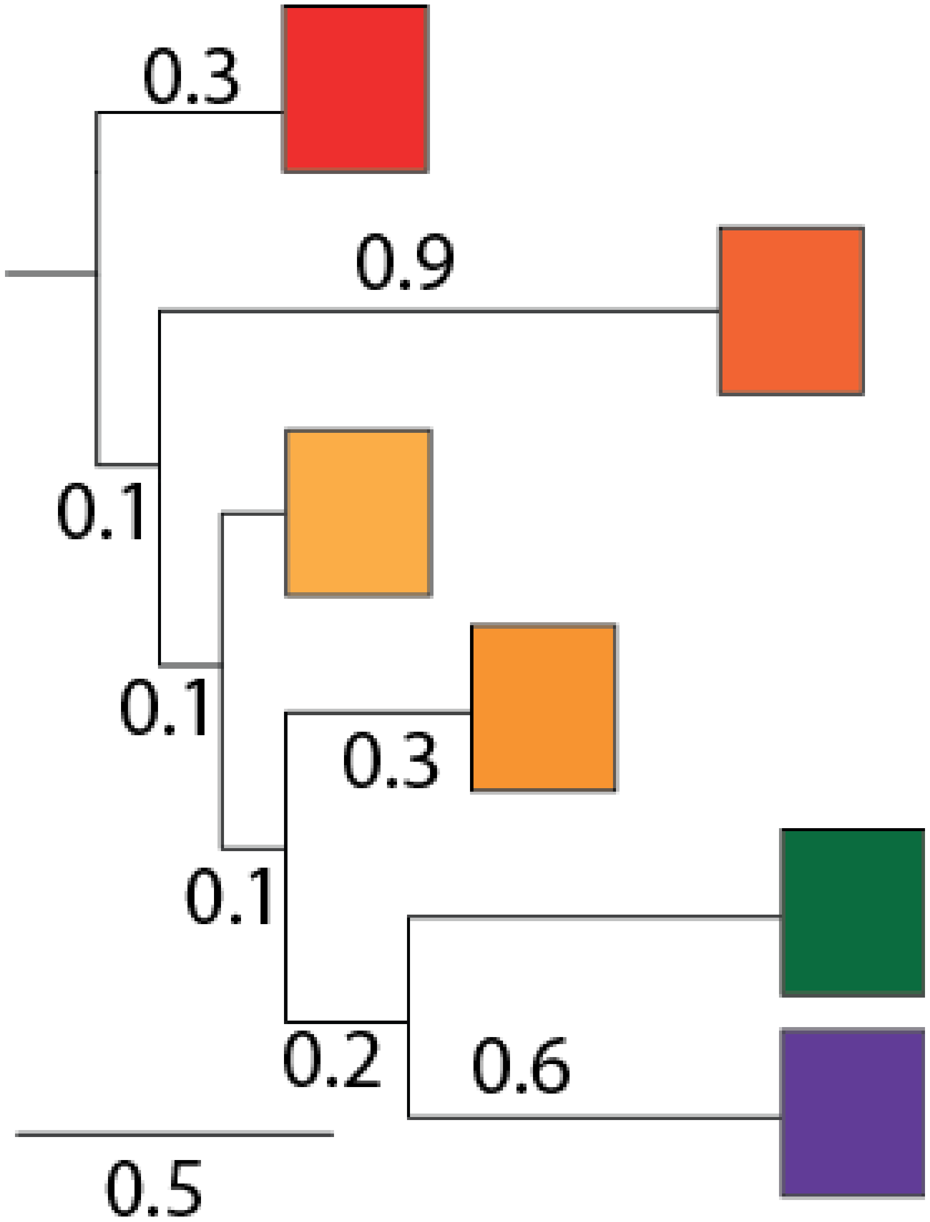
- Ecology proposes mechanisms for genetic variation between populations
- Epidemiology studies the patterns, causes and effects of disease on populations
- An ecological study is an epidemiological study in which the unit of analysis is a population rather than an individual



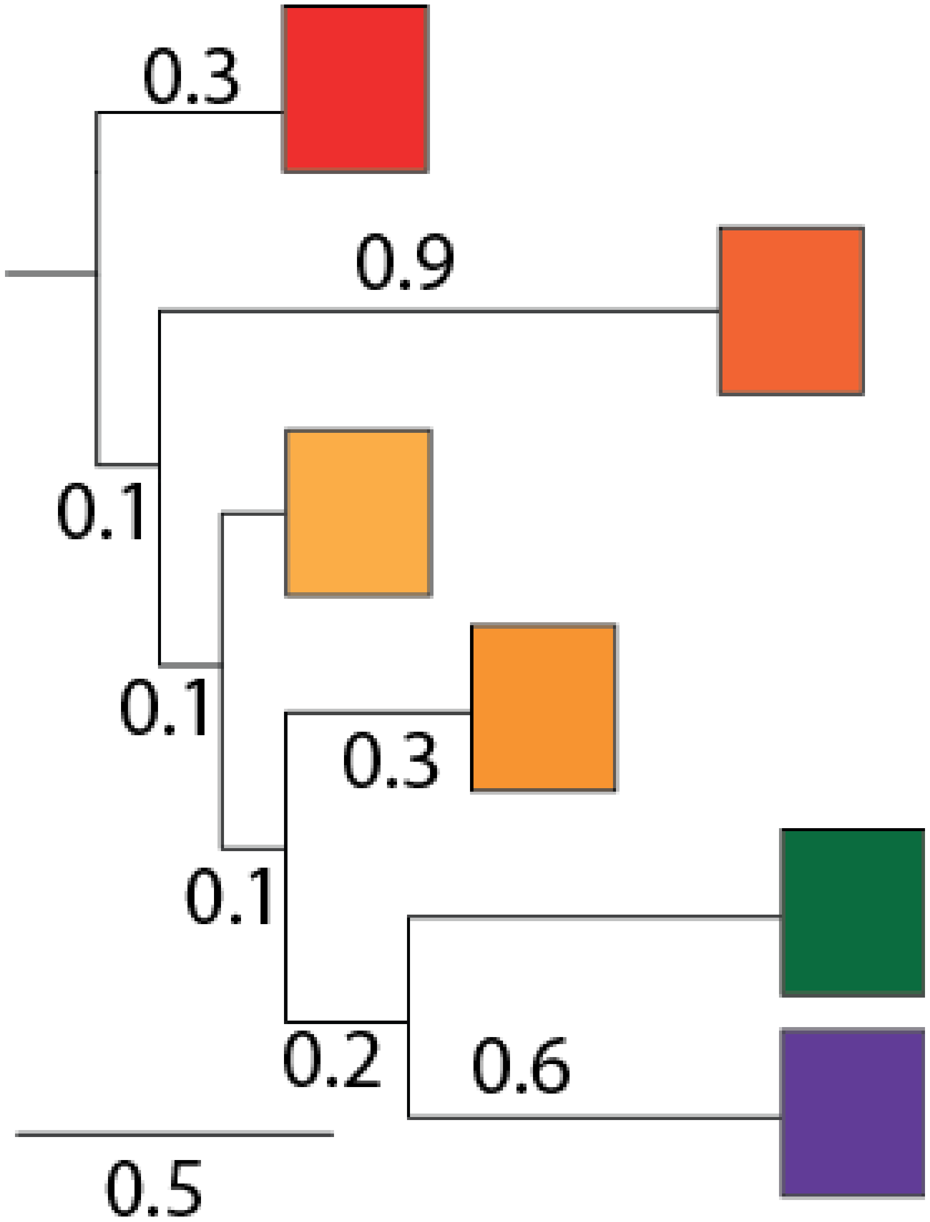
Our Toy Tree



Dimensions

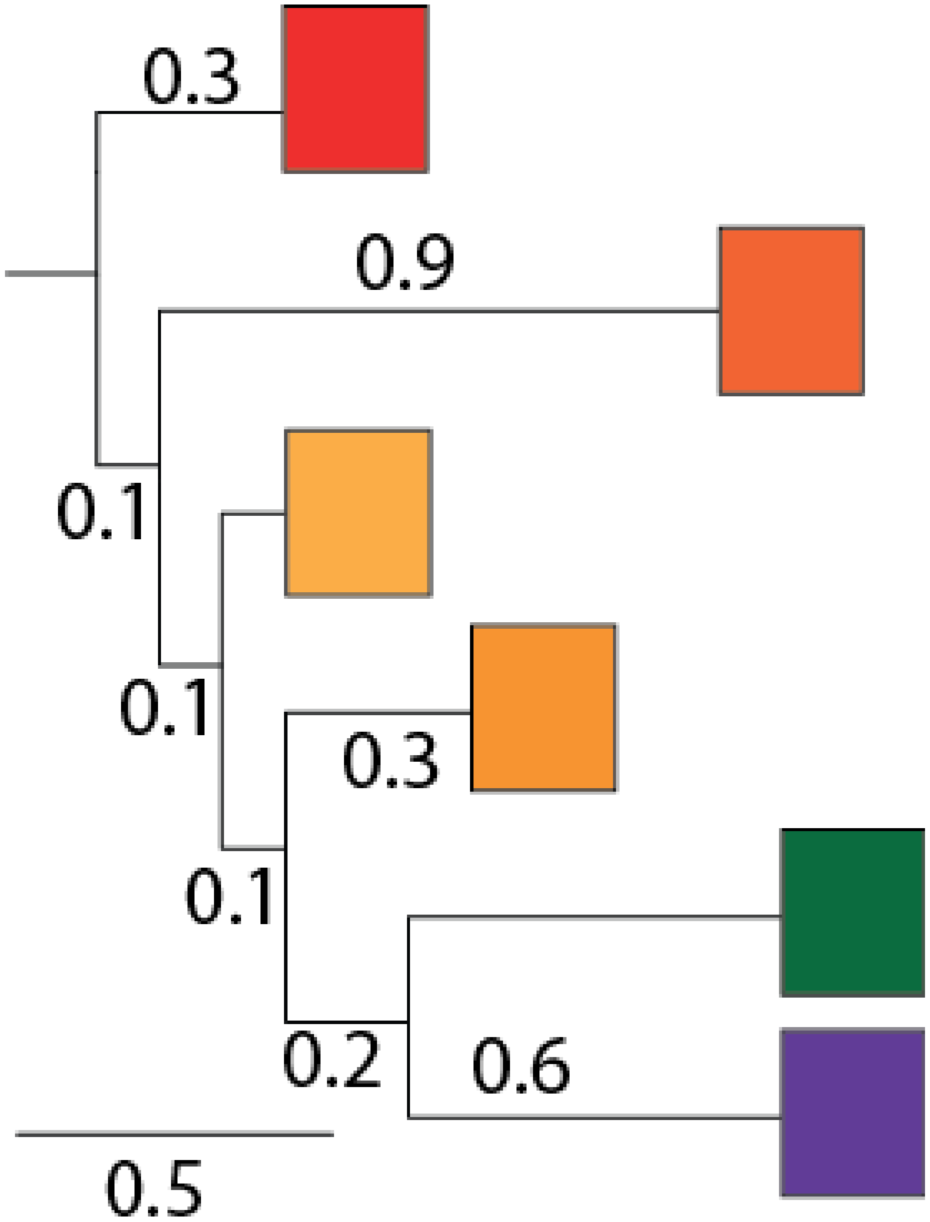


Dimensions



Vertical dimension has no meaning

Dimensions

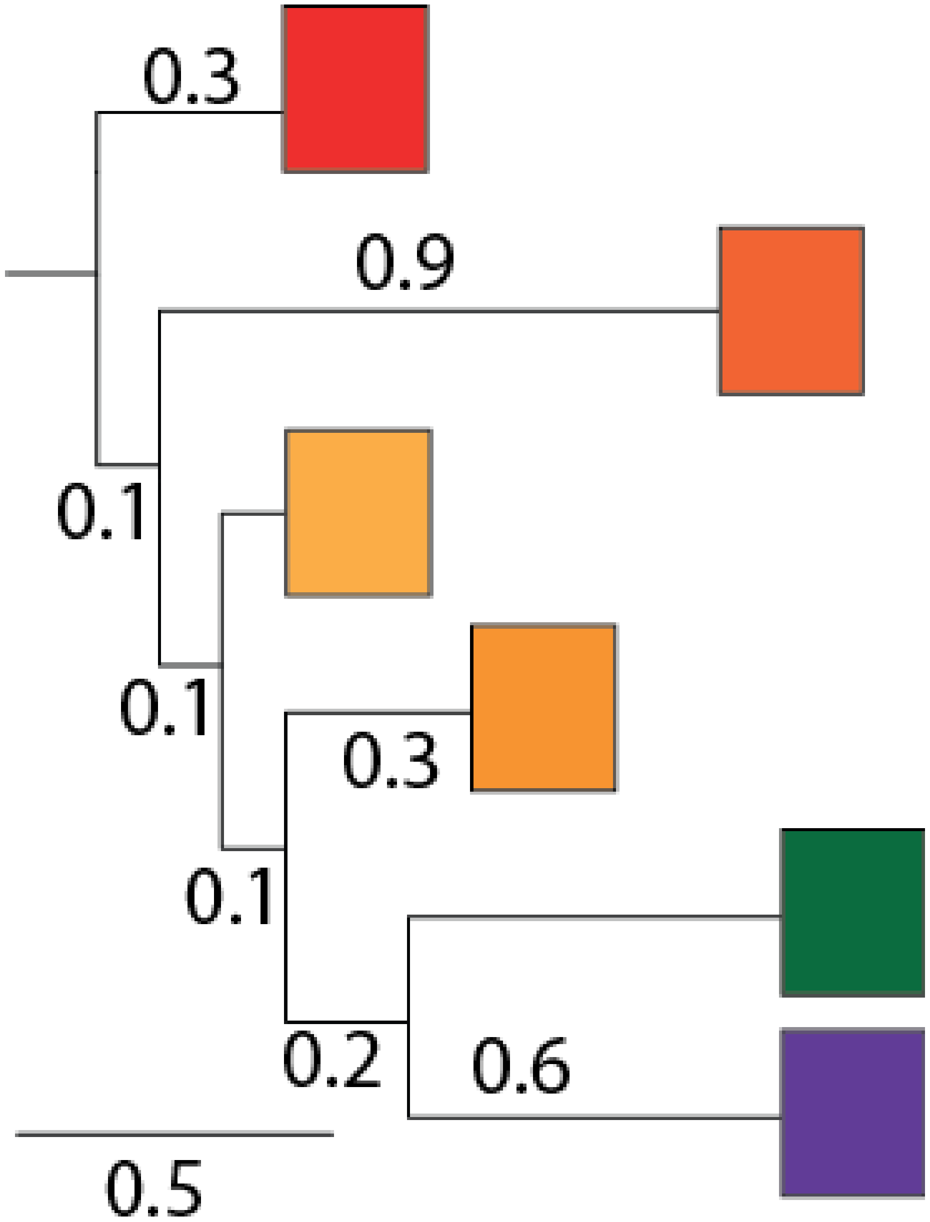


Vertical dimension has no meaning

Horizontal dimension gives the amount of genetic change

Horizontal lines show evolutionary changes in lineages over time (i.e. branch lengths are informative)

Dimensions



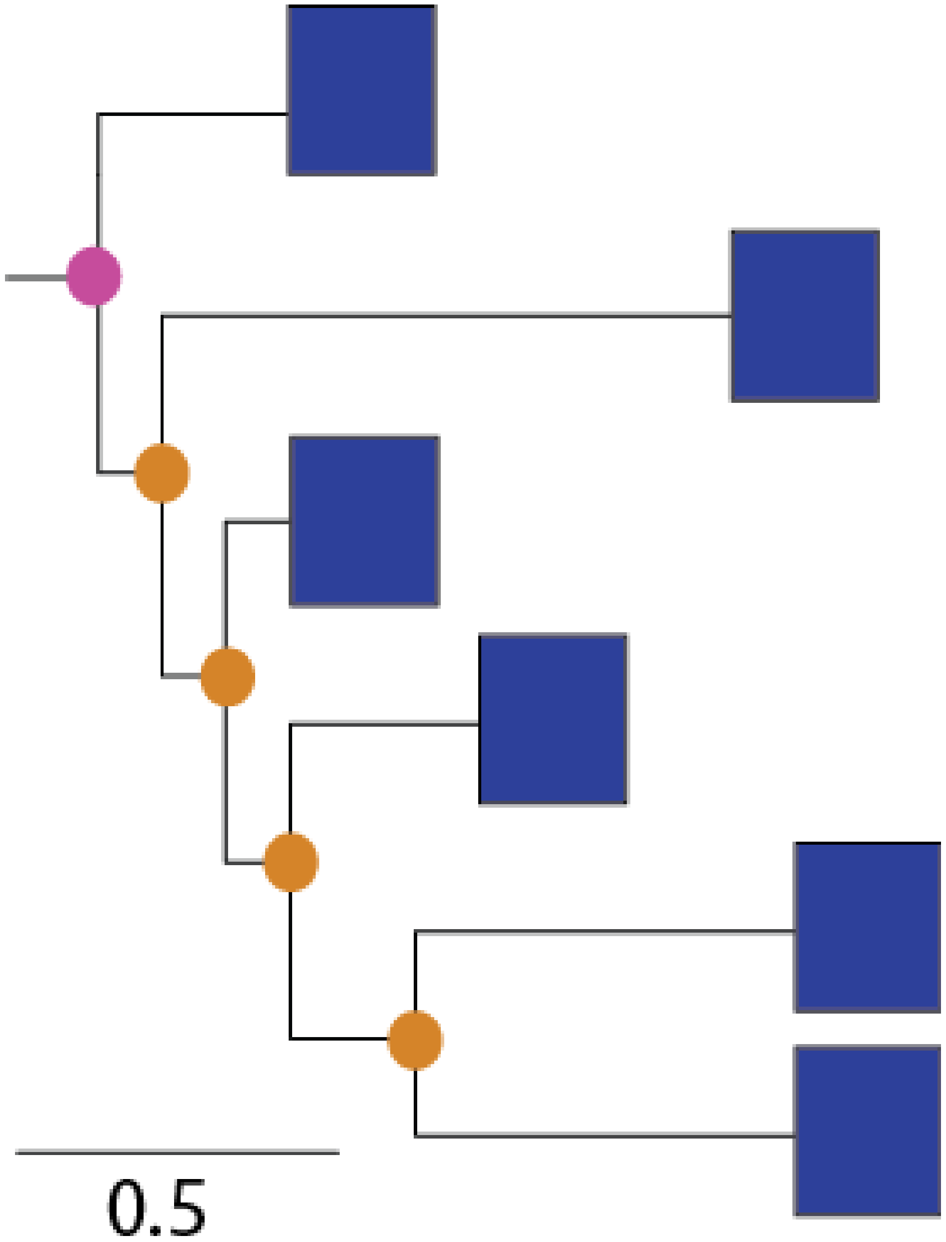
Vertical dimension has no meaning

Horizontal dimension gives the amount of genetic change

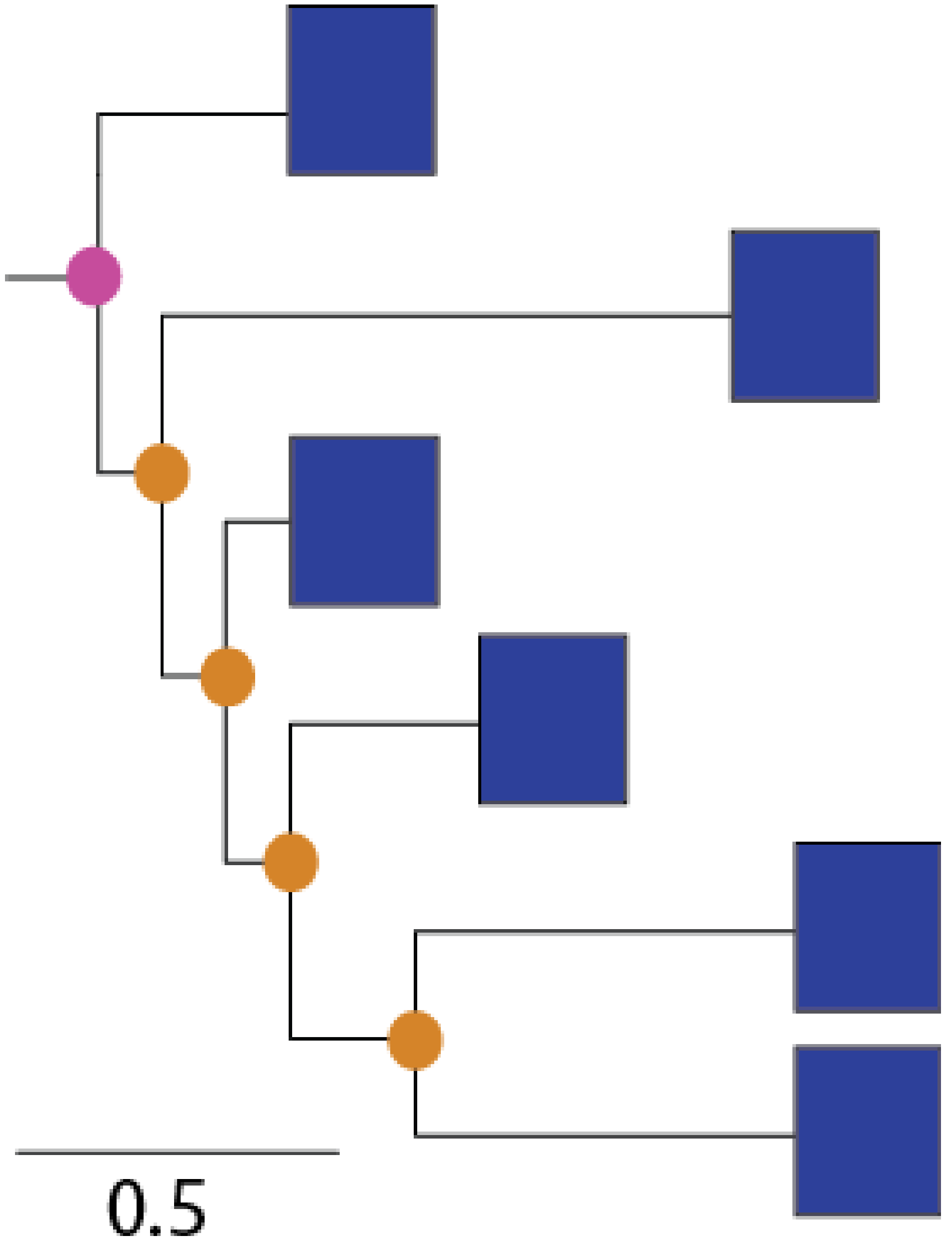
Horizontal lines show evolutionary changes in lineages over time (i.e. branch lengths are informative)

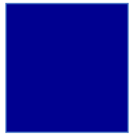
Scale bar shows the length of branch equivalent to 0.5 nucleotide substitutions/site

Structure



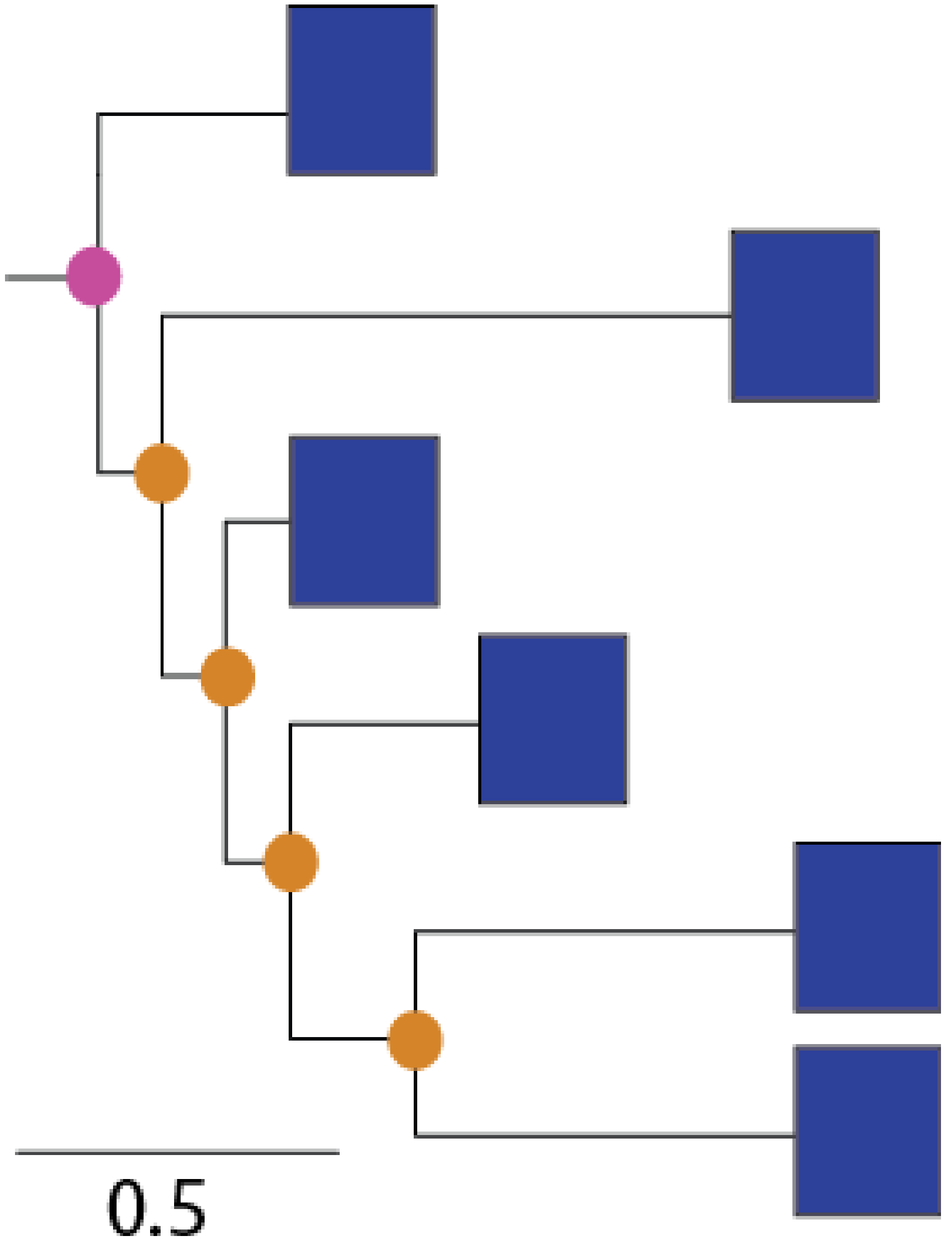
Structure





Tip/Taxon/Virus/Leave
-Sequenced Viruses
-we also know date and location of isolation, host and clinical features

Structure



Tip/Taxon/Virus/Leave
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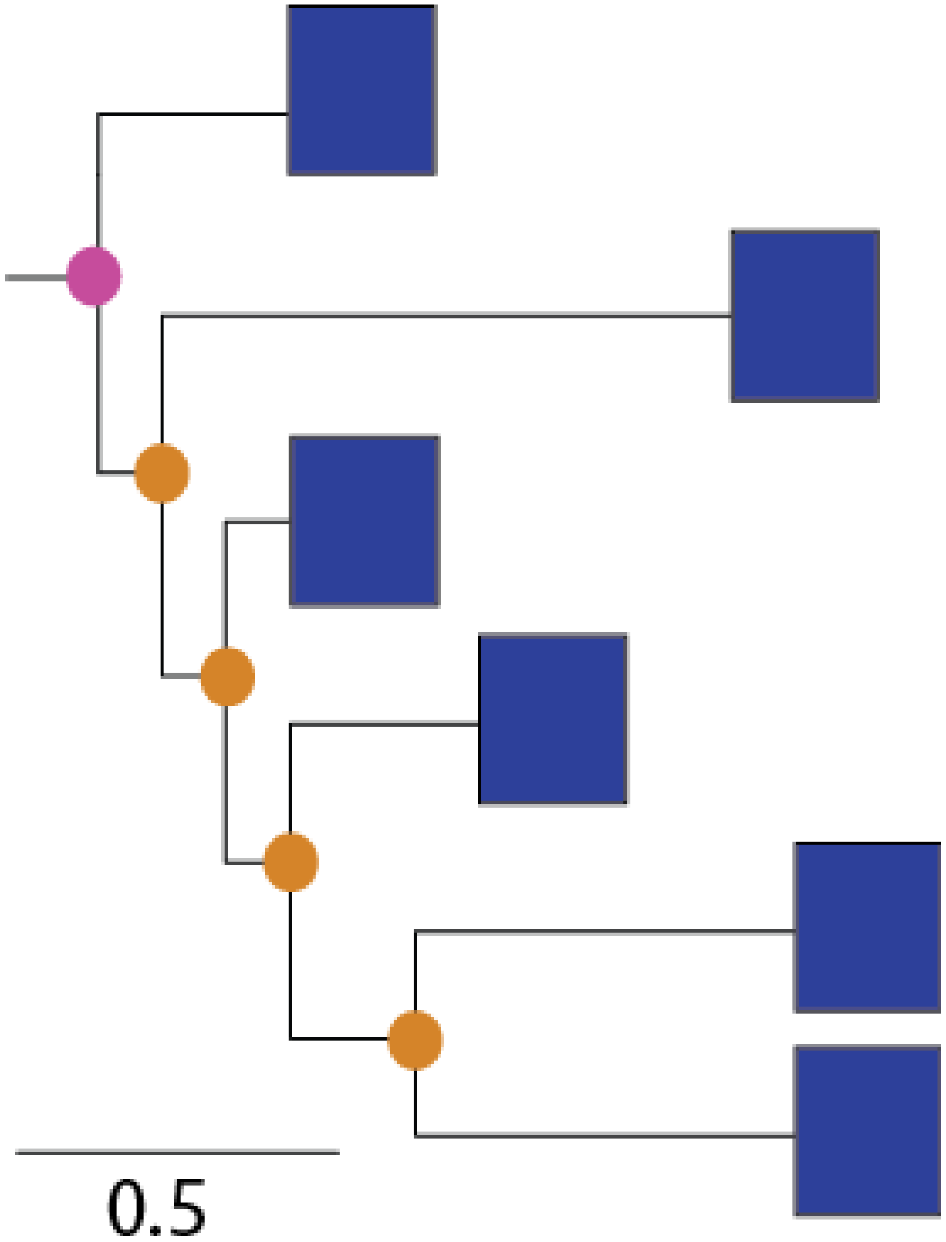
Internal Node
Putative common Ancestor

Structure

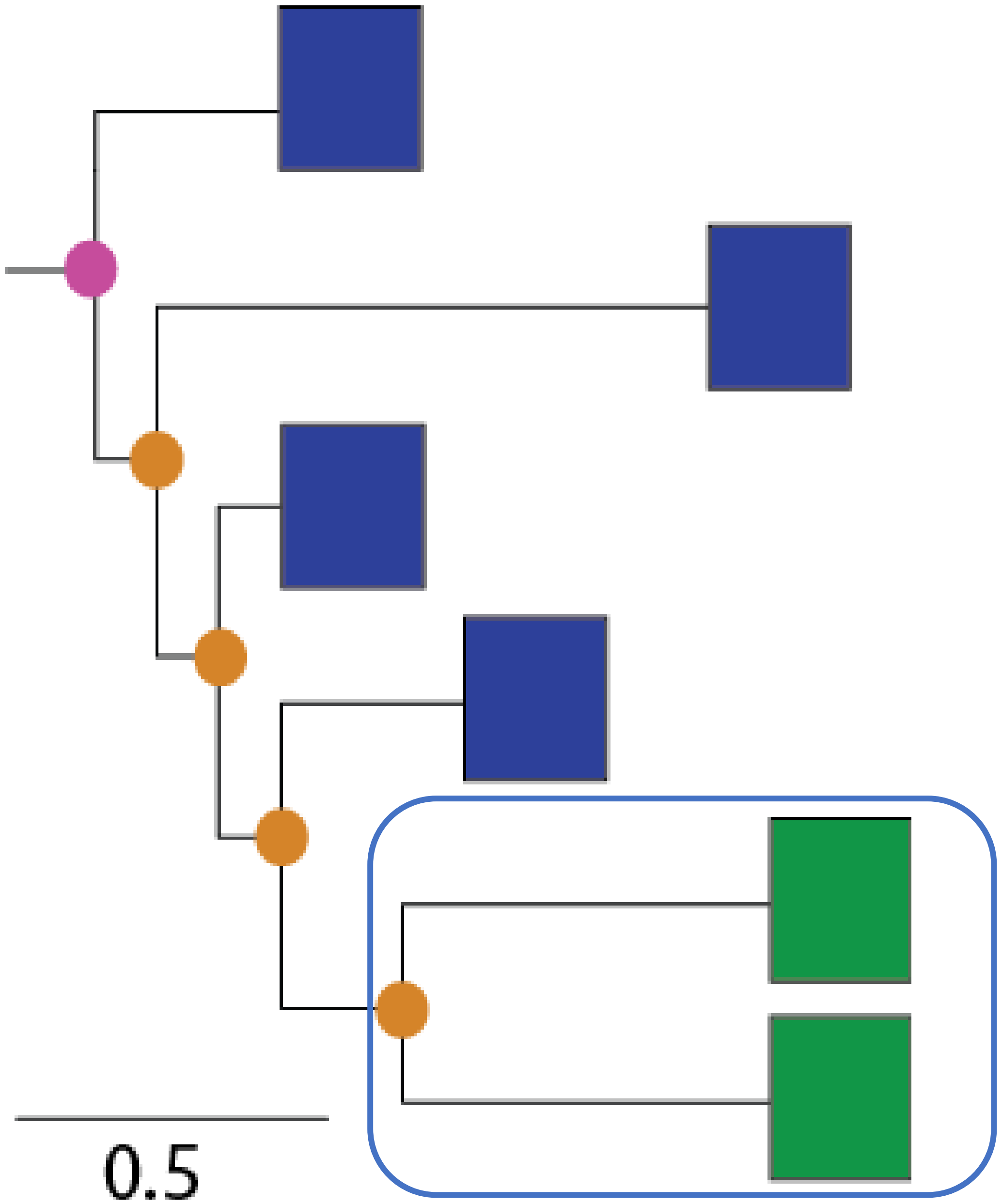


Tip/Taxon/Virus/Leave
-Sequenced Viruses
-we also know date and location of isolation, host and clinical features

Internal Node
Putative common Ancestor

Root – provides an order for sequence of events leading to observation (sequenced virus)

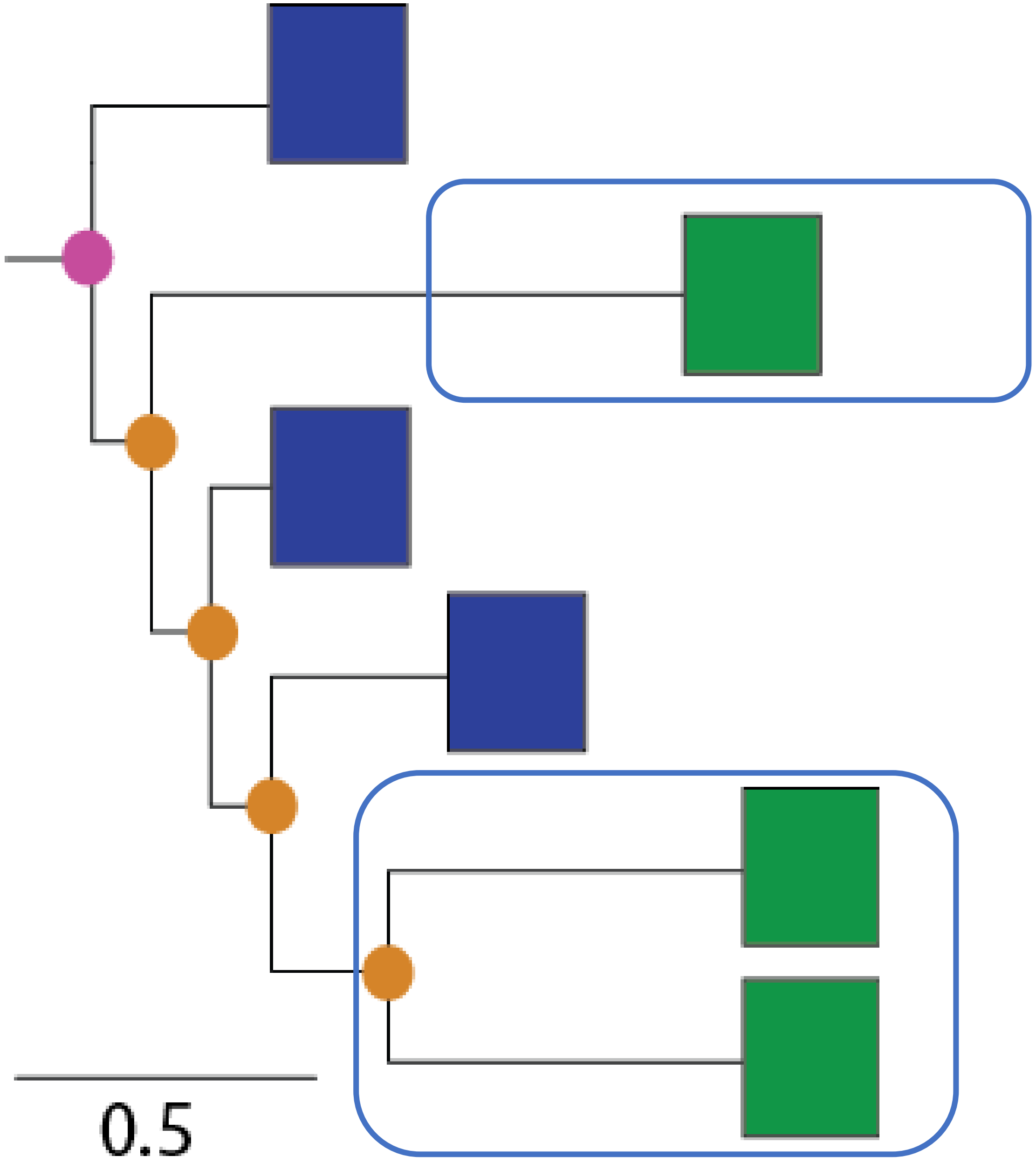
Terminology



Monophyletic group (Clade) = all members share common ancestry

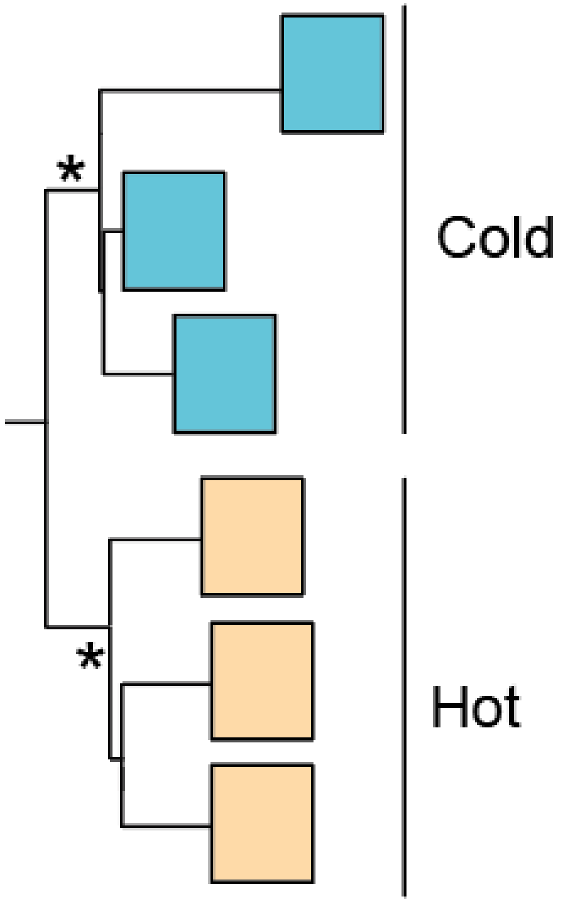
Jargon dictionary

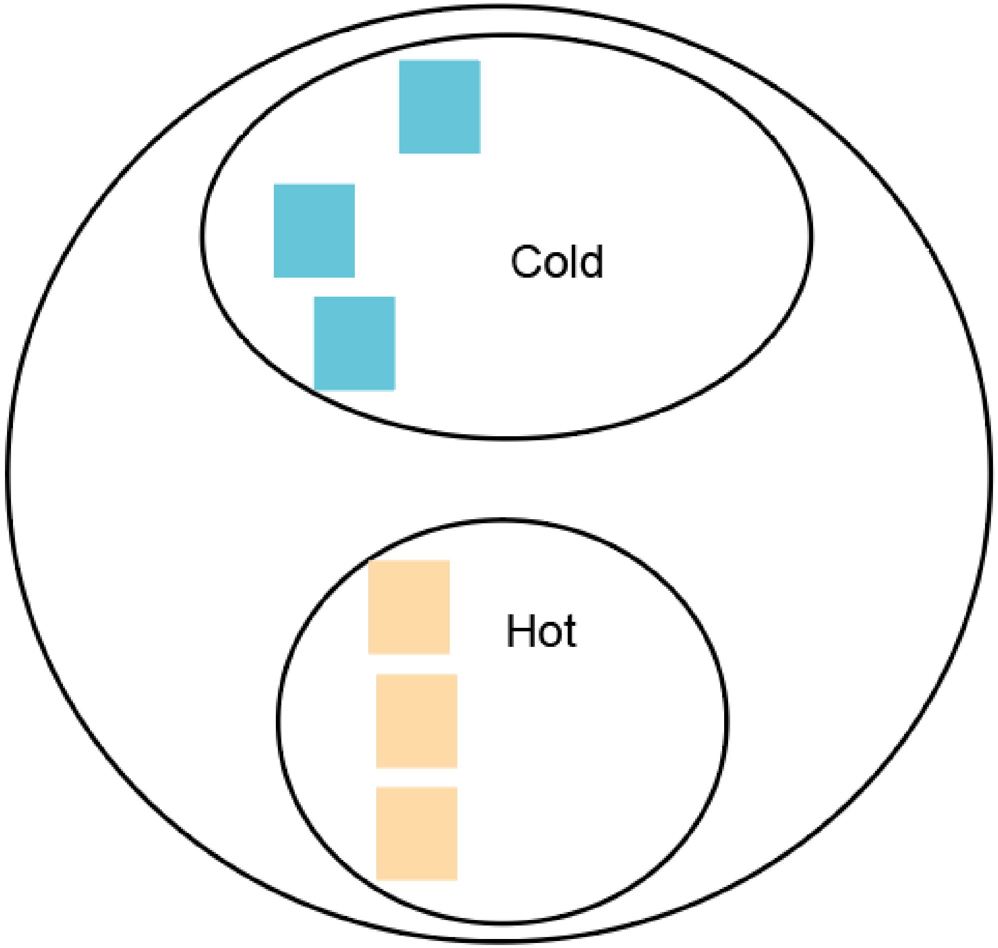
Terminology



Paraphyletic = a group of organisms descended from a common ancestor, but not including all the descendant groups

Reading a tree: Relationships





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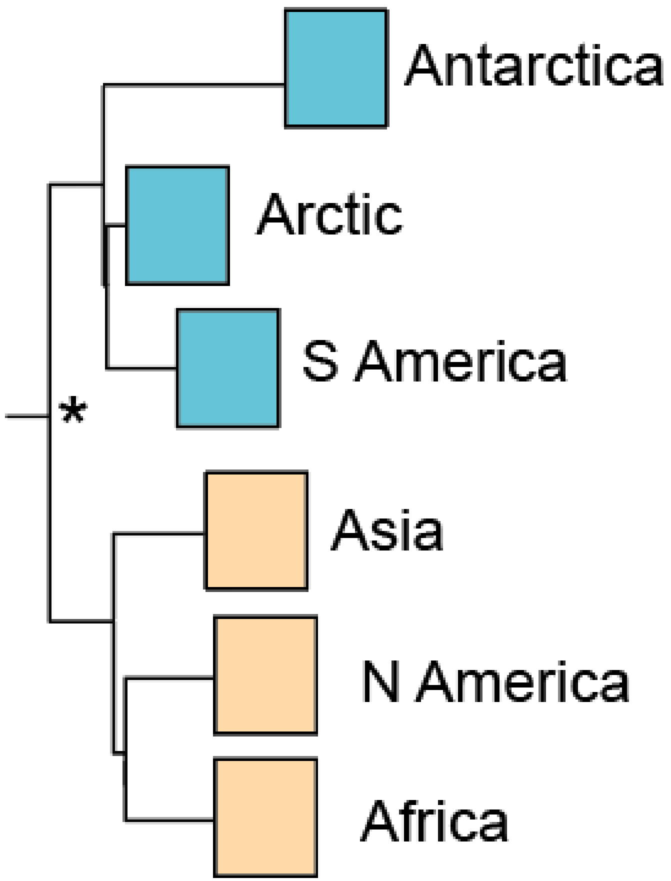

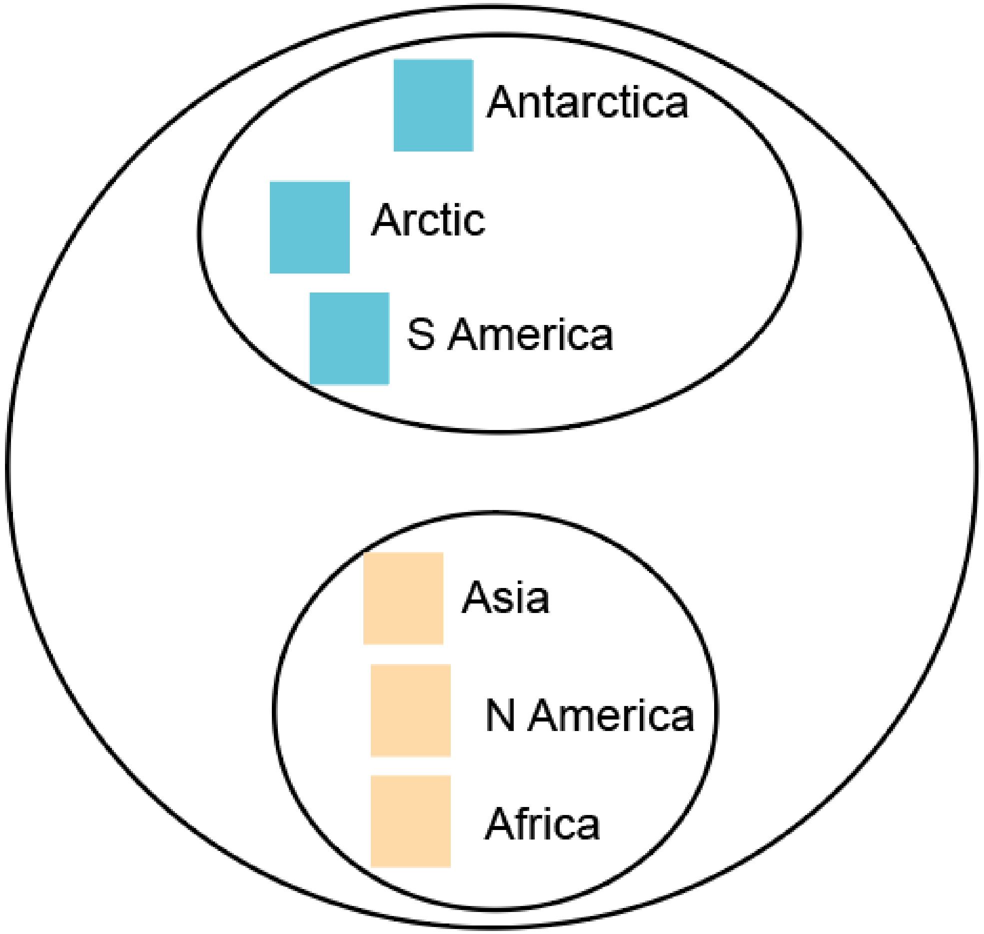

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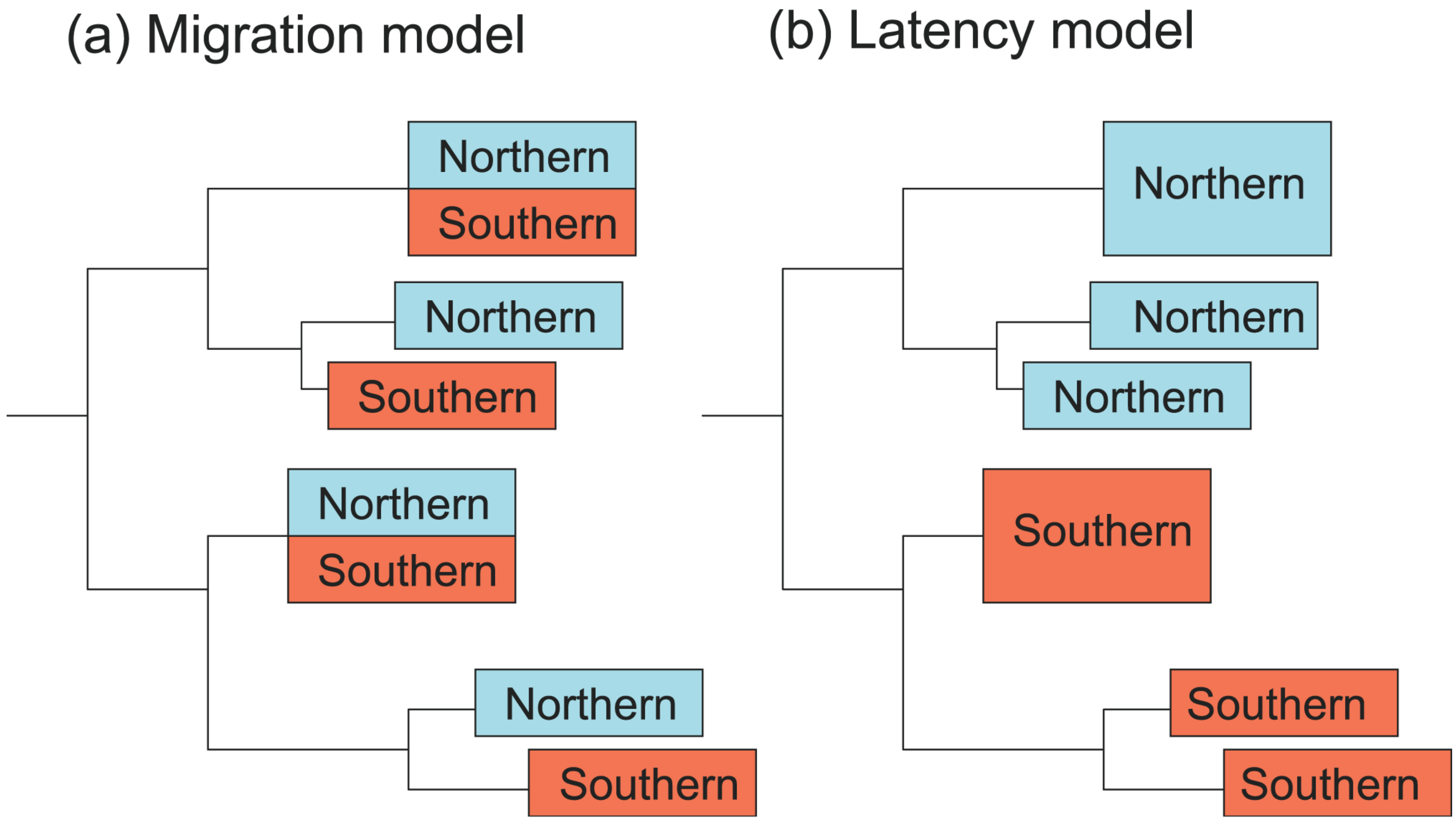
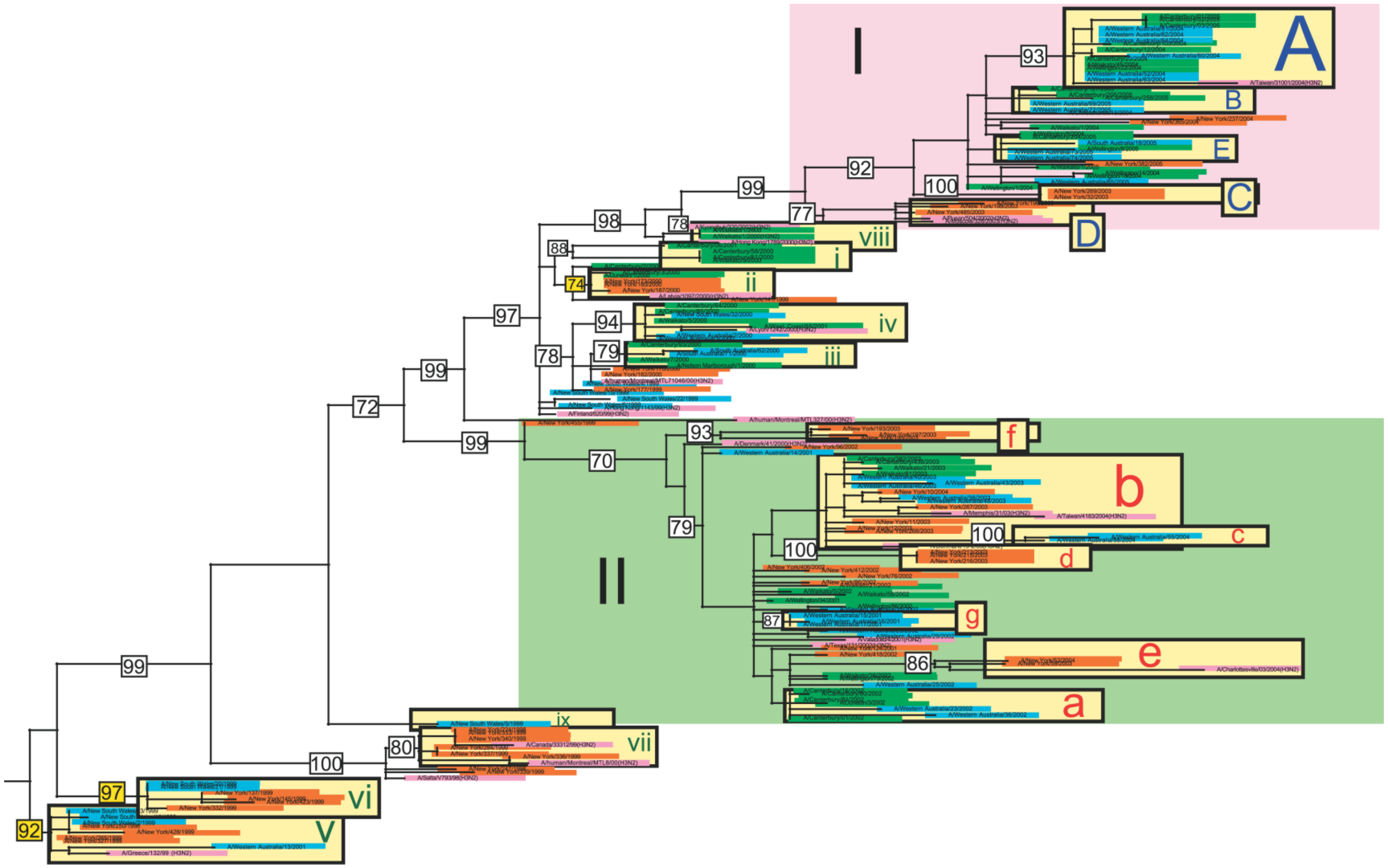


Nelson *et al* (2007) PLoS Pathogens https://doi.org/10.1371/journal.ppat.0030131

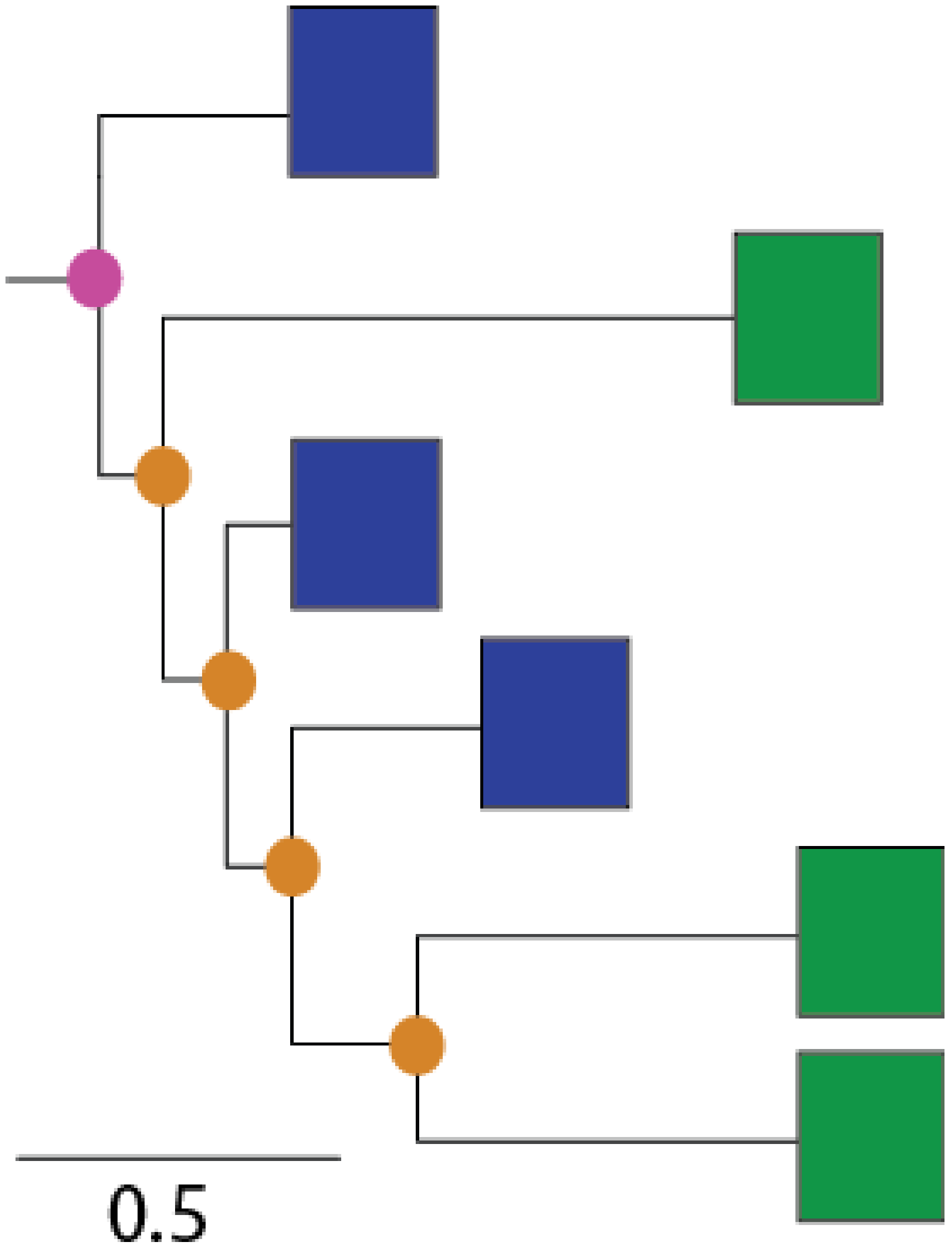
Reading a tree: Relationships

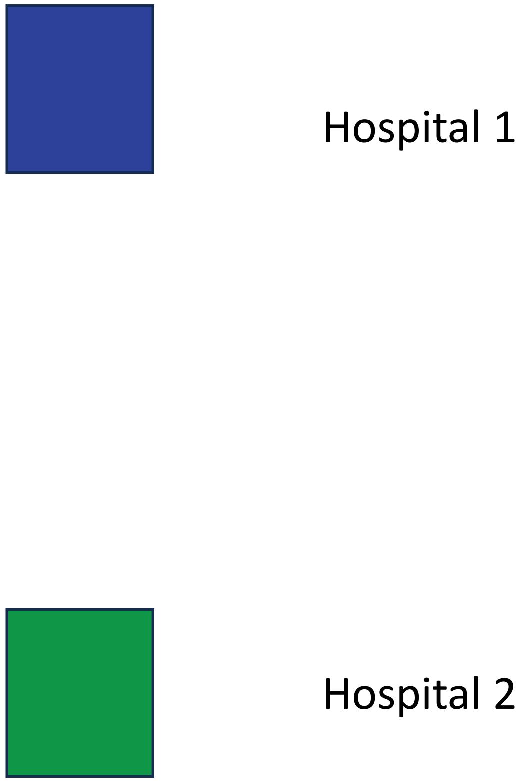


Influenza A virus migrates between hemispheres during non-epidemic periods, rather than persisting locally at low levels during the influenza "off-season".

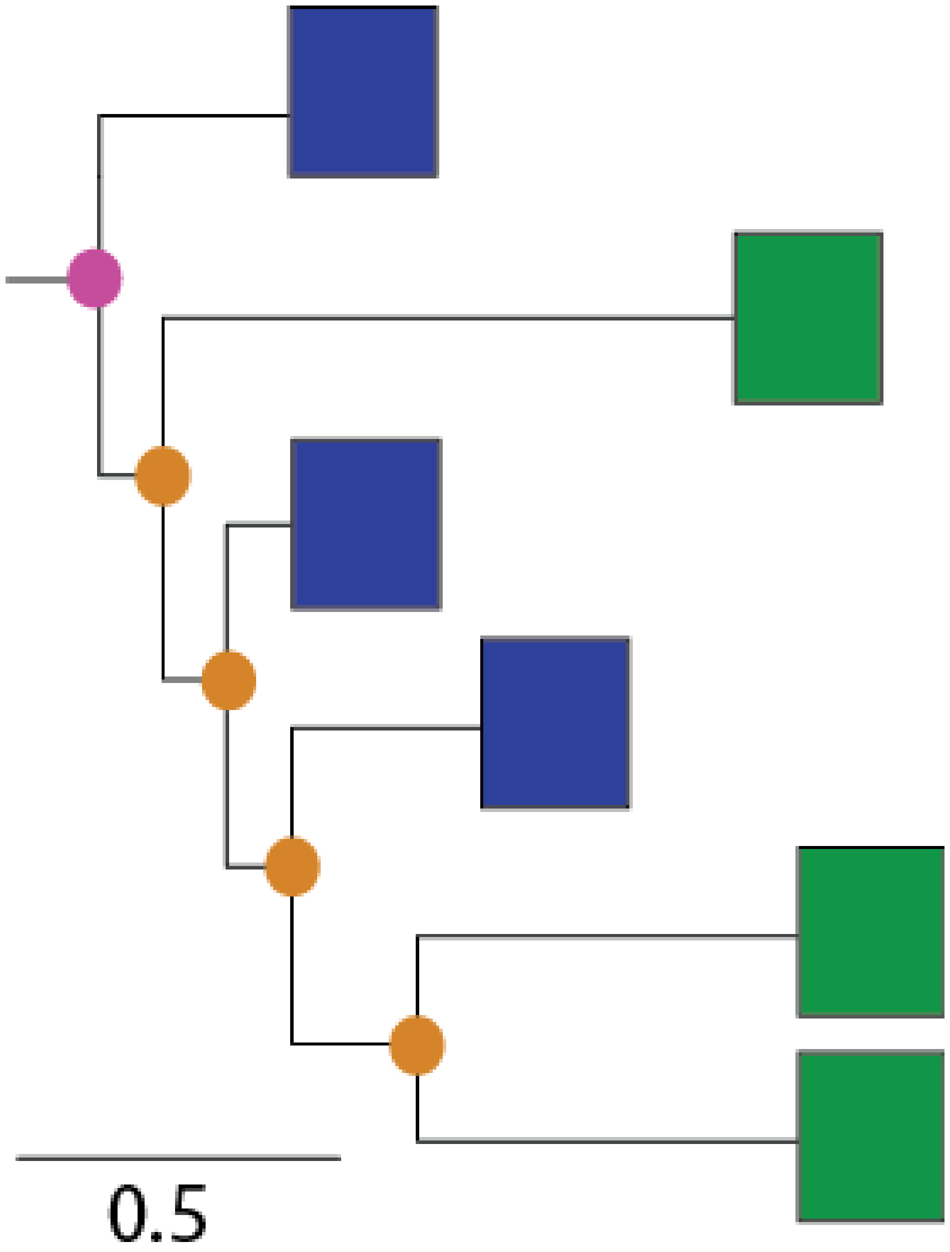
Nelson *et al* (2007) PLoS Pathogens https://doi.org/10.1371/journal.ppat.0030131

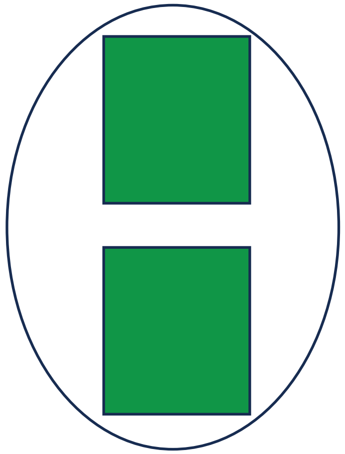
Reading a tree: Relationships



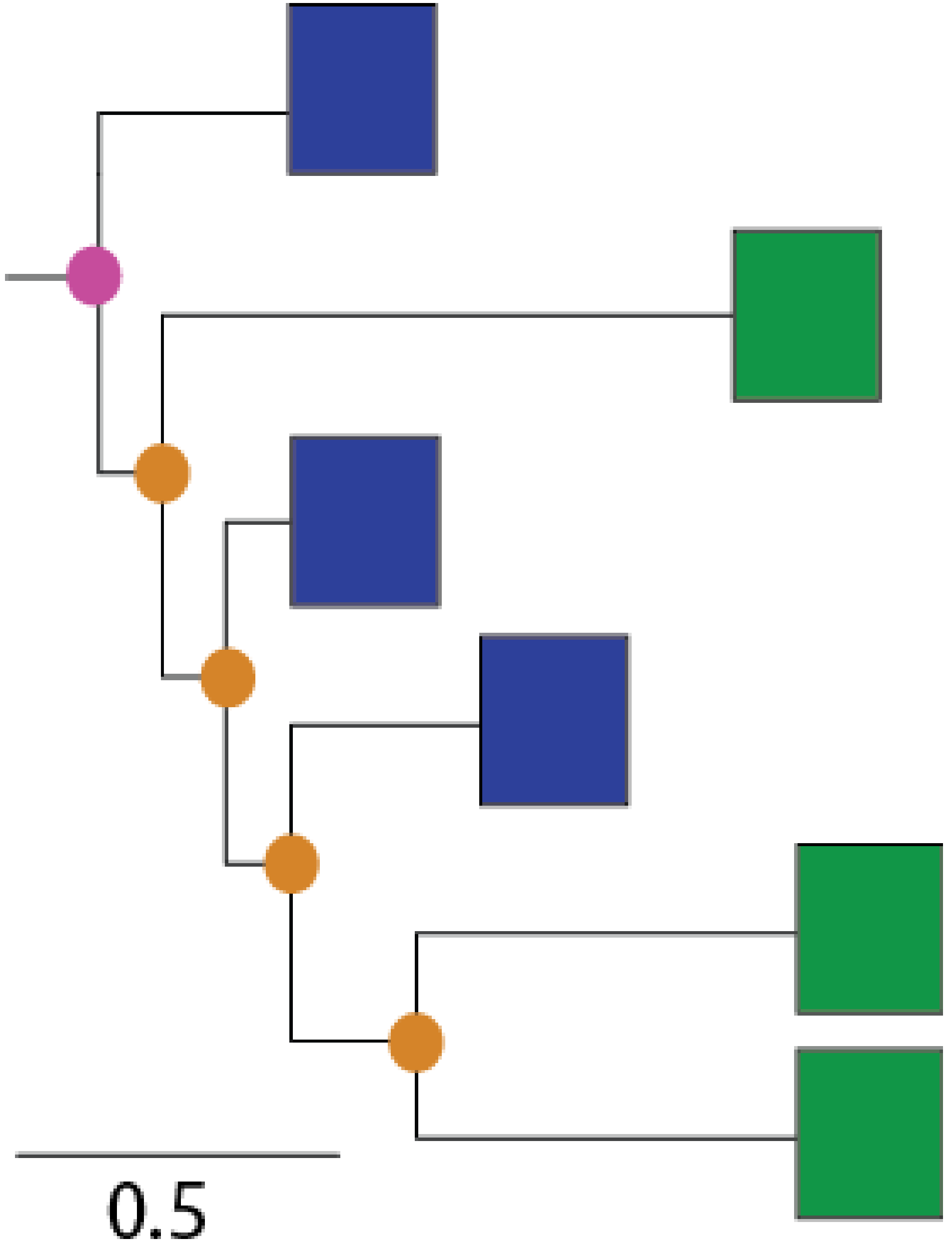


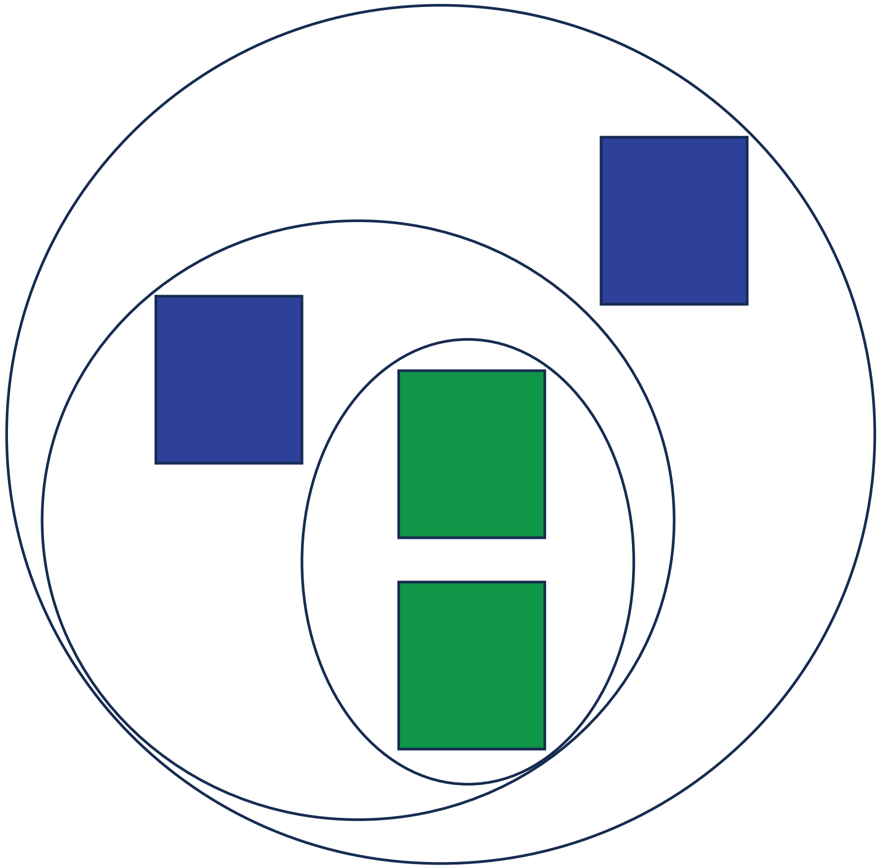
Reading a tree: Relationships



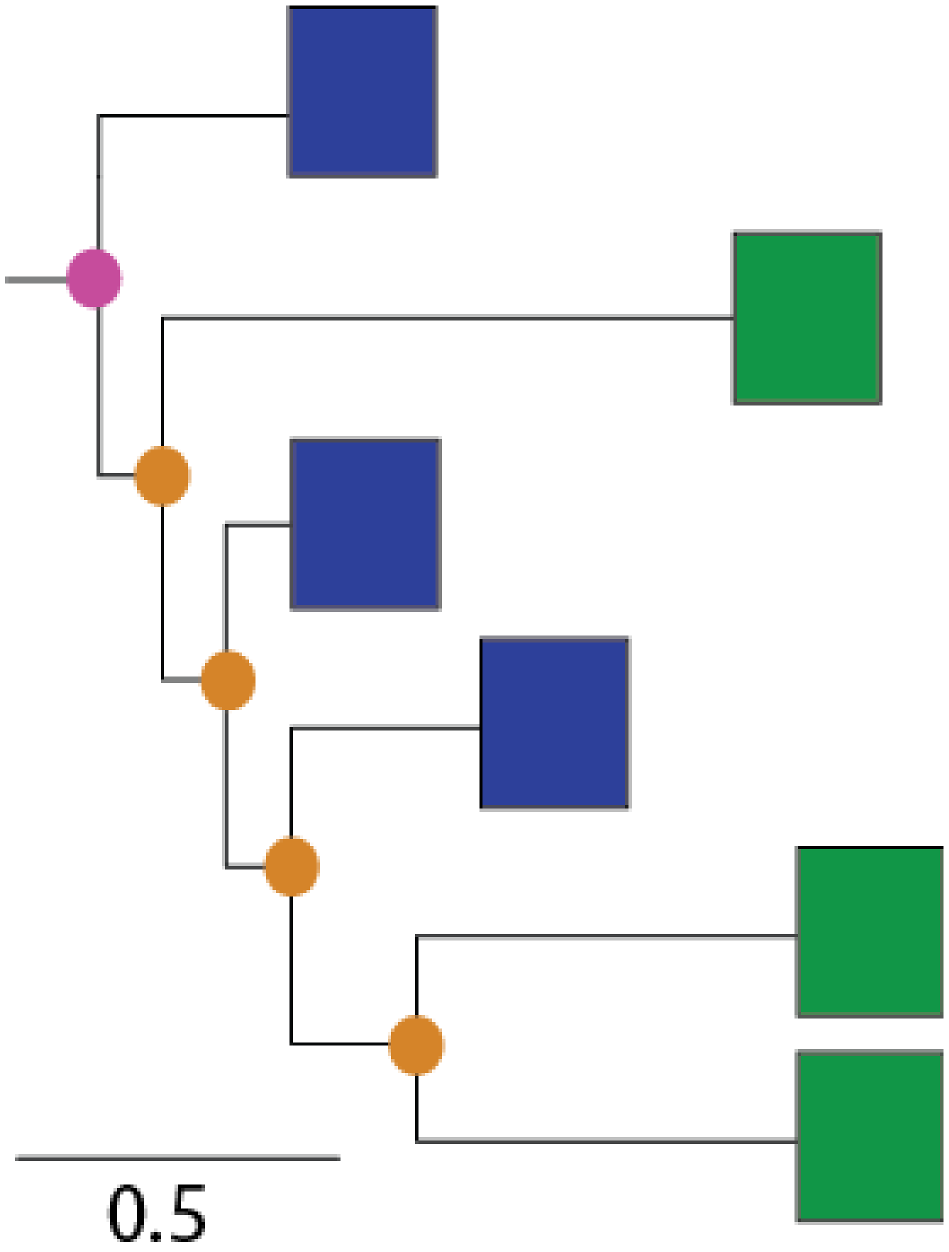


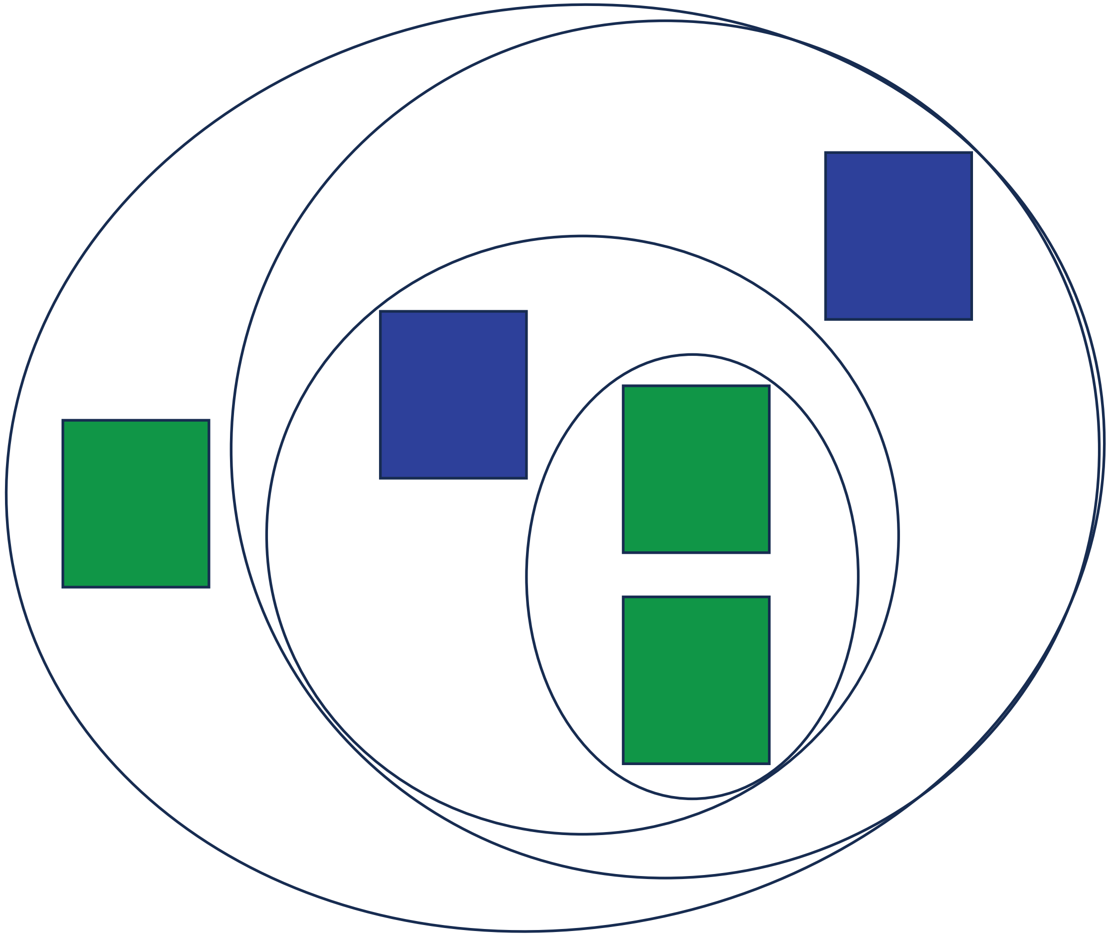
Reading a tree: Relationships



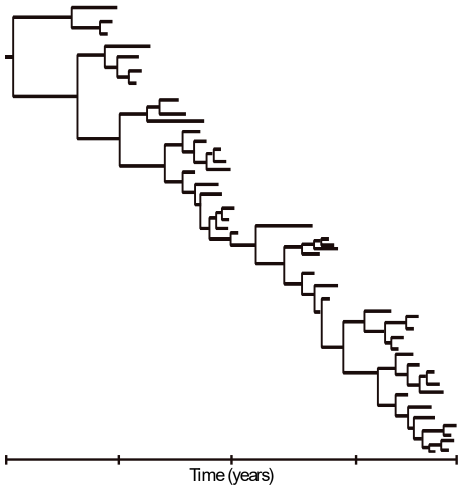


Reading a tree: Relationships

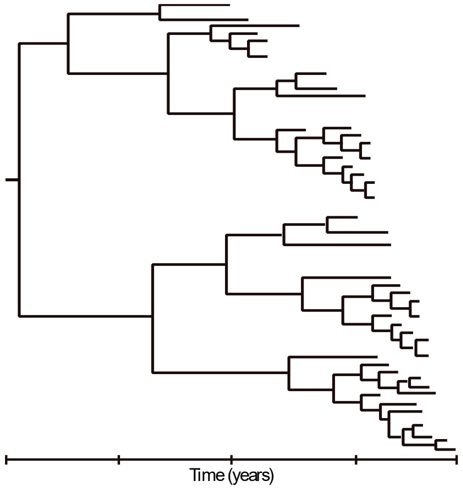




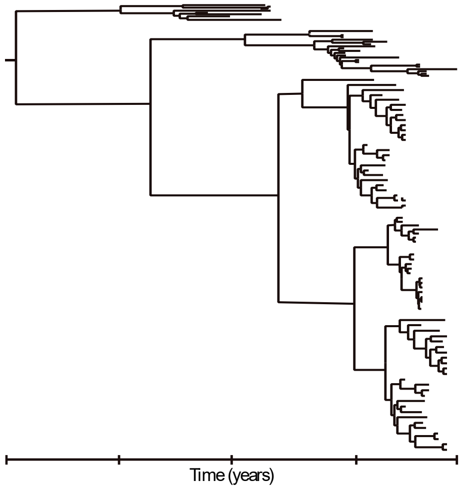
Phylogeny is shaped by host ecology and epidemiology



Human Influenza (Seasonal)



Avian Influenza (Wild bird)



Avian Influenza (Domestic poultry)

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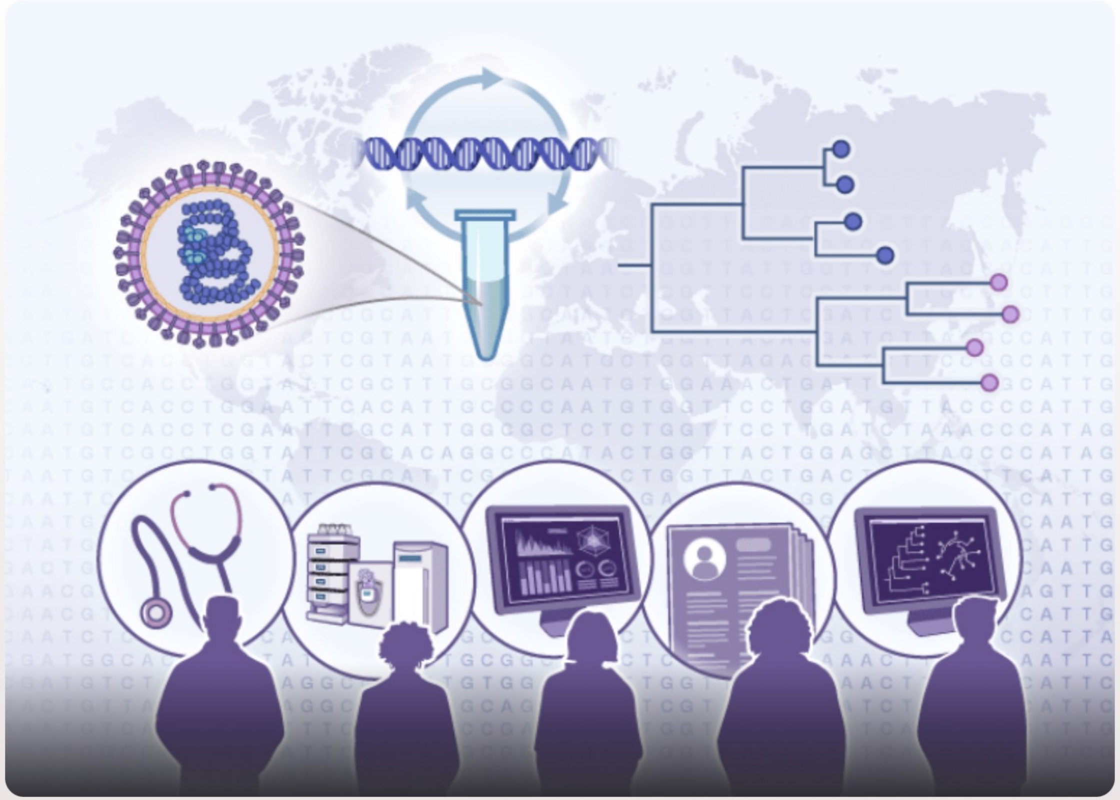
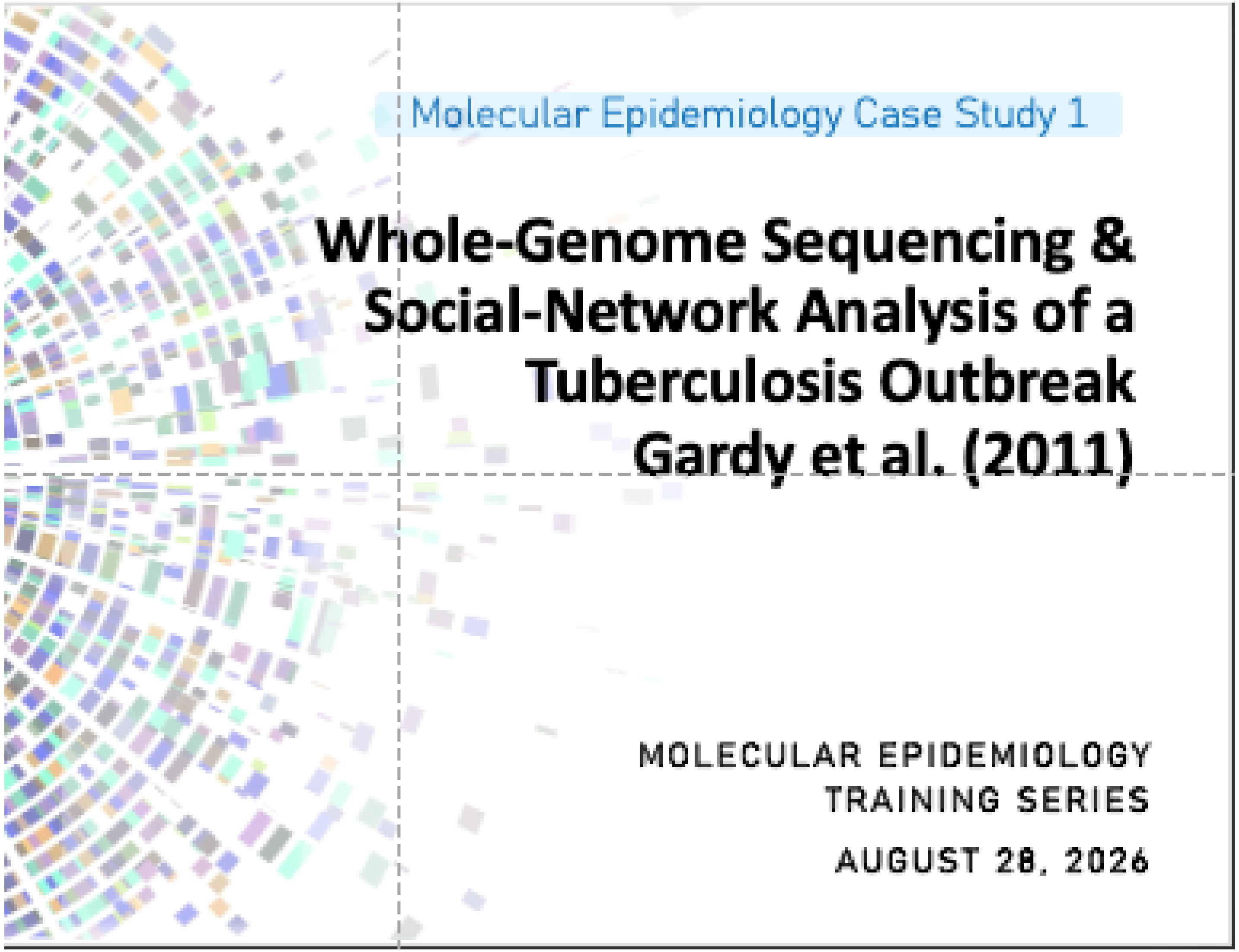
Pathogen Genomics for Public Health

Harness the power of pathogen genomics in infectious disease response.



https://learn.hms.harvard.edu/pathogen-genomics-public-health

Next week-
Case study



Please Read!

- For each case study, there will be an article that participants should plan to read before the session. The article has been provided along with a reading guide in the materials folder on GitHub (provided in the Welcome email- https://github.com/StaPH-B/southeast-region/tree/master/trainings/2026%20ELC%20Training).

Epidemic Process

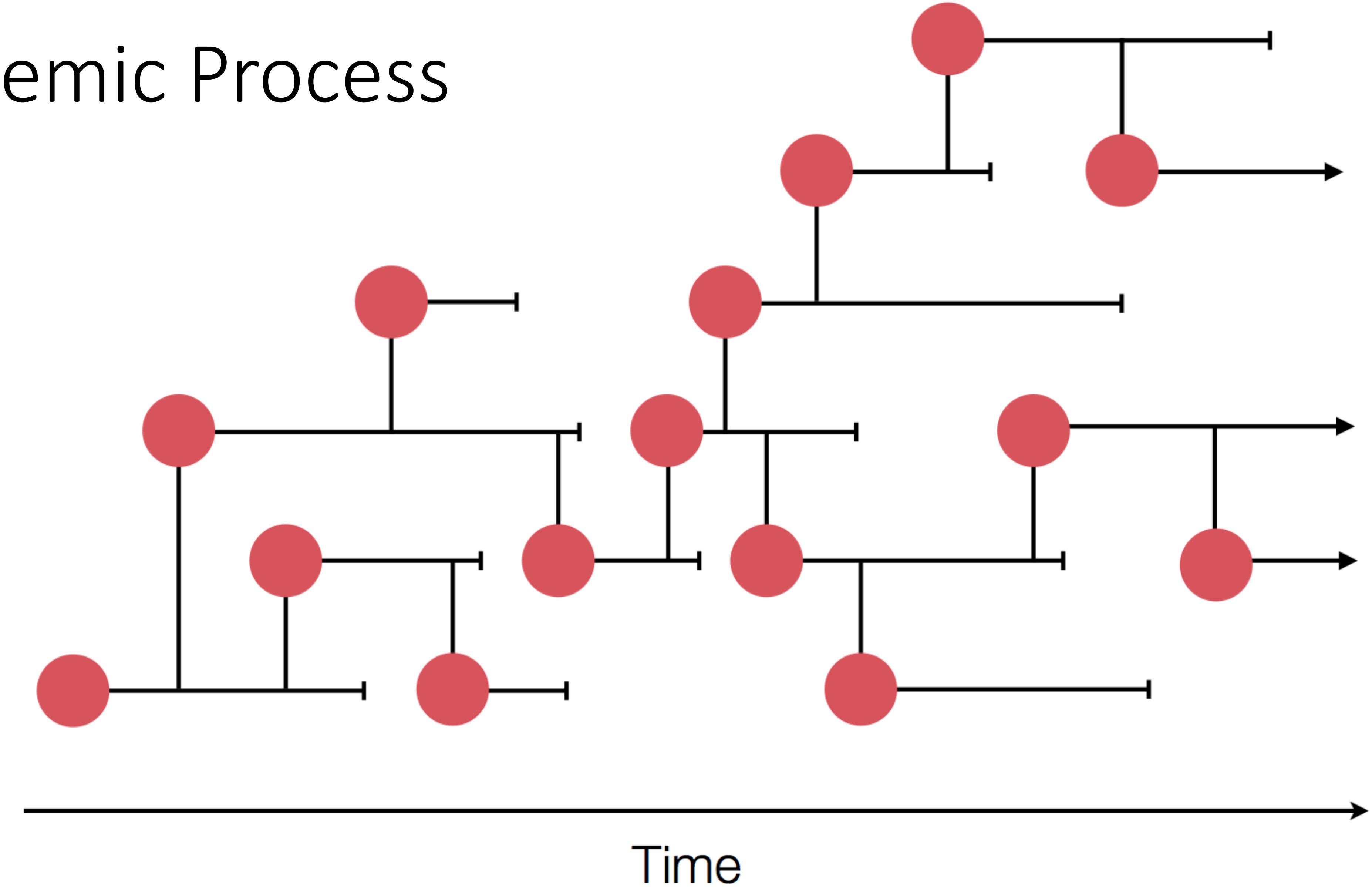
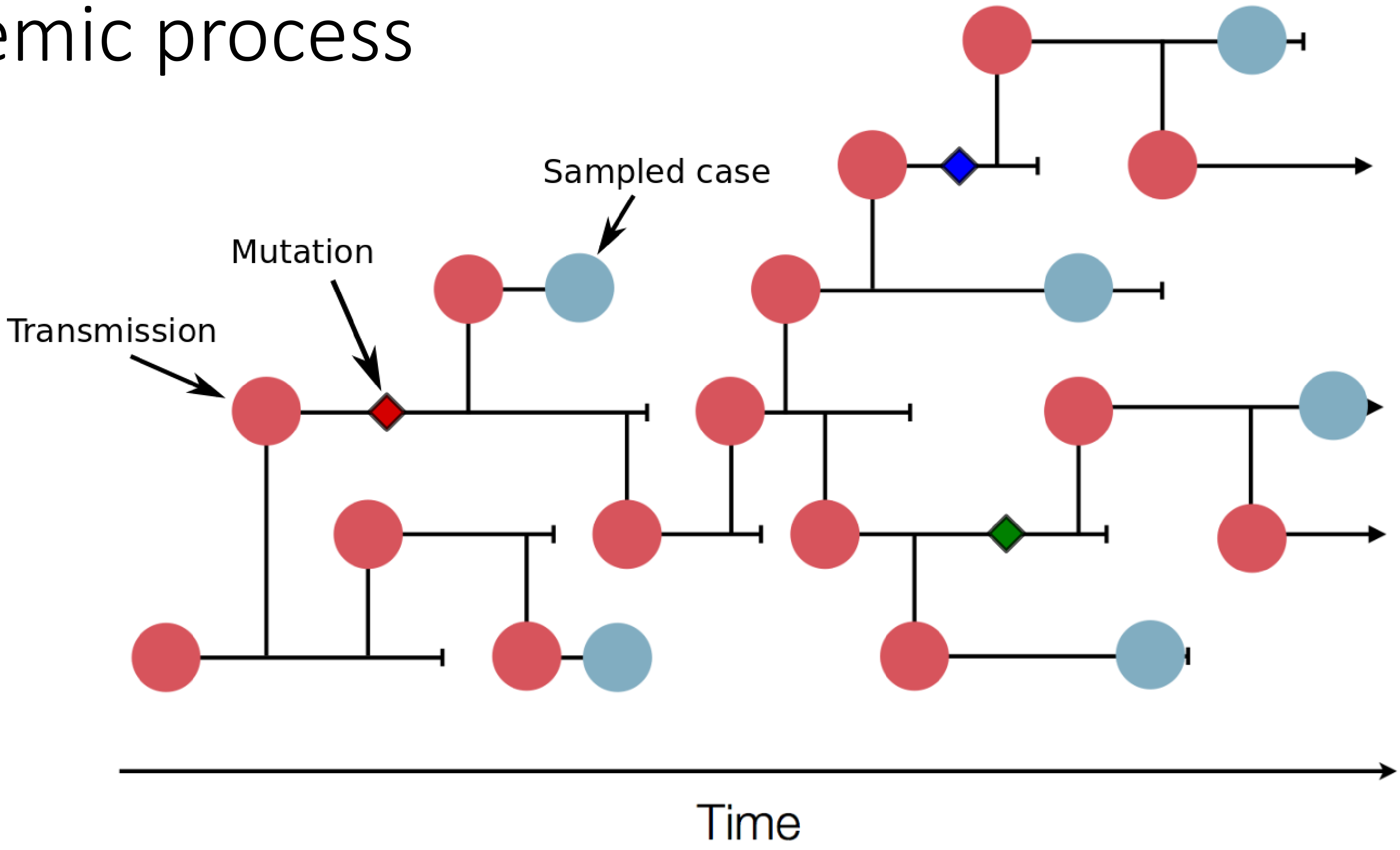
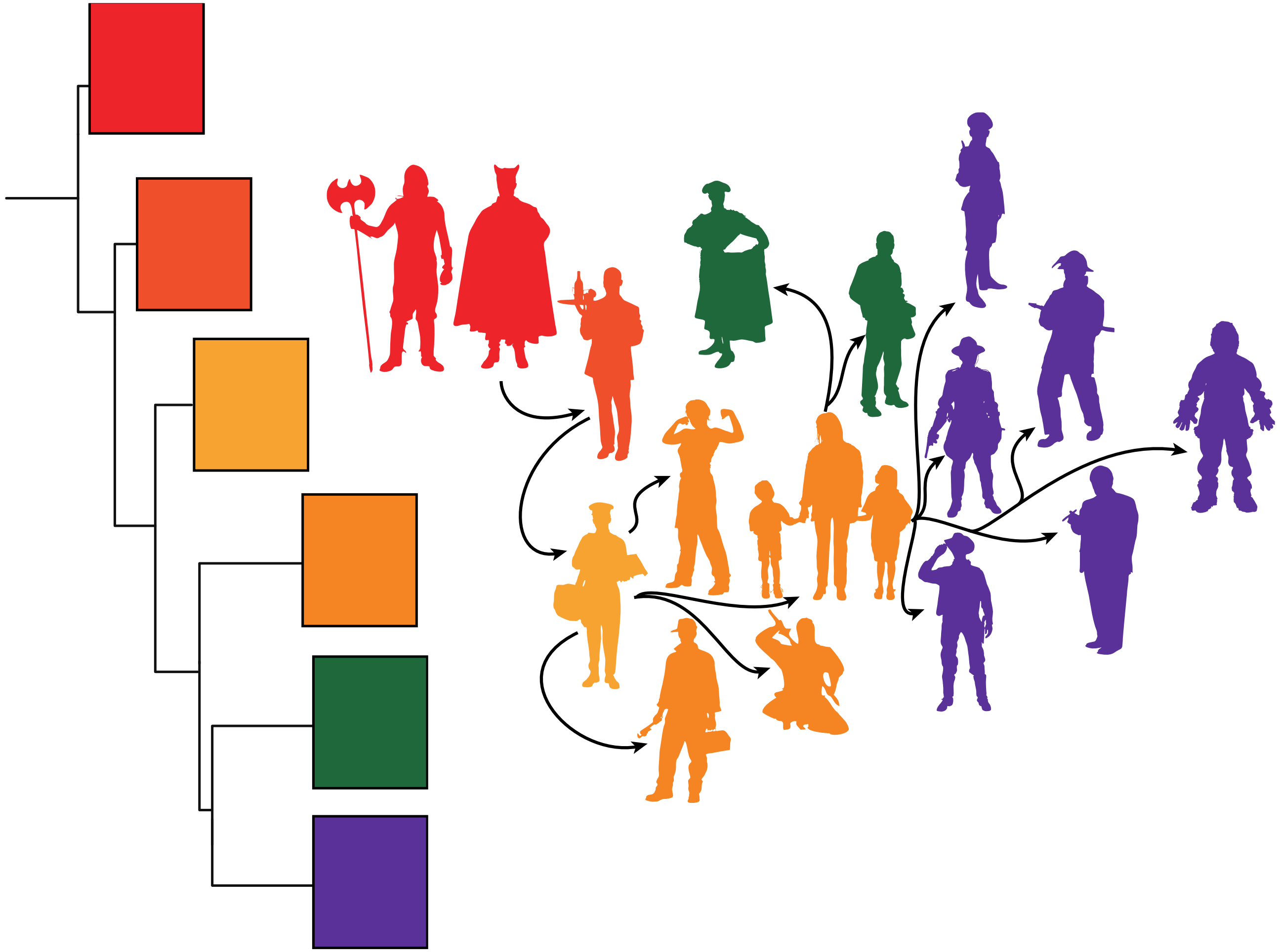


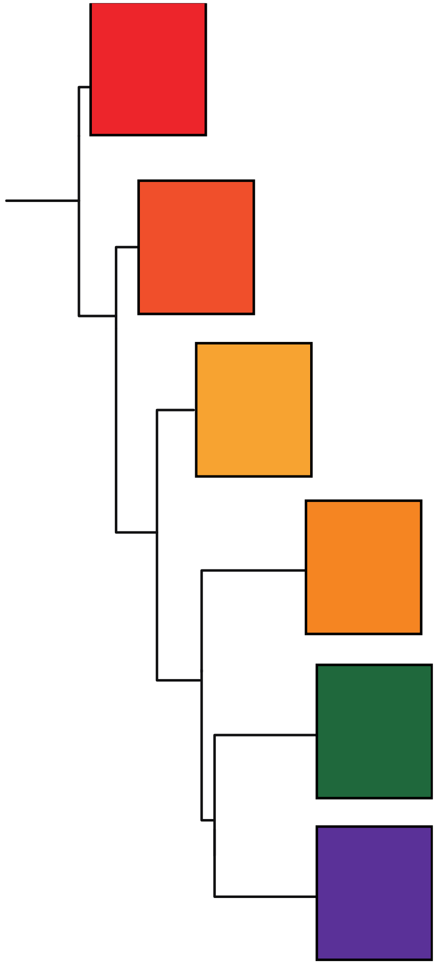
Image from Trevor Bedford: https://bedford.io/projects/phylodynamics-lecture/intro.html#/6

Viruses accumulate mutations during the epidemic process





Phylodynamics



The rapid evolution of many pathogens, particularly RNA viruses, means that their evolution and ecology occur on the same timescale, and therefore must be studied jointly to be fully understood.

- All observations (i.e. viruses isolated during an outbreak) are related by common descent
- Evolution is bifurcating (Tree-like)
- small genetic distances = closely related

Data integration and statistical modeling